Modelling Financial Markets

Chapter 17: Bubbles and Crashes

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Opening: Can Statistics Spot a Bubble Before It Bursts?

> *"The second theorem of the financial instability hypothesis is that over periods of prolonged prosperity, the economy transits from financial relations that make for a stable system to financial relations that make for an unstable system."*
>
> — *Minsky (1992)*

- Minsky: a long period of stability shifts borrowers towards speculative and Ponzi finance
  - Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic Kindleberger & Aliber (2005)
- S&P Composite price-dividend ratio, 1871–2026: GSADF (Generalised Sup Augmented Dickey–Fuller) $= 3.41$ vs the 95% critical value 2.39
  - the full-sample ADF (Augmented Dickey–Fuller) test (-1.11) sees nothing
- After an industry price run-up above 100%, a 40% crash followed in 45% of cases Greenwood, Shleifer & You (2019)
  - against 11% for a random industry-month; 53 episodes in 27 industries
  - the mean return over the next 2 years is not significantly different from zero: run-ups raise the risk, not the predictable loss
- The credit behind the boom: after the R-zone (rapid growth of credit and asset prices), a household-sector crisis begins within 3 years in 34% of cases
  - against 4% in normal periods; only 25 crises in the data

▸ **Central chart: S&P Composite (Shiller) Price-Dividend Ratio, 1871–2026**

Can Statistics Spot a Bubble Before It Bursts?

Bubbles, Crashes and Their Detection

- Prices sometimes rise far faster than any plausible change in fundamentals
  - Then they lose most of their value
- Questions
  - can statistics detect such episodes **in real time**?
  - can it say **when** they end?
- Part I
  - What a bubble is, run-ups and crashes in the data
  - Tests for explosive prices and their date-stamping
- Part II: the LPPLS model (Log-Periodic Power Law Singularity)
  - Regime switching, crypto and artificial-intelligence (AI) stocks
  - Two case studies, one pass of the AI-assisted research loop

What you should be able to do

- derive the rational-bubble condition and separate the fundamental value from the bubble, with what can be tested
- compute GSADF and the BSADF date-stamping on rolling windows, with valid critical values
- compute an LPPLS fit and interpret the stability of its critical time and its alarms
- interpret booms and busts with a Markov-switching model and the test of the number of regimes
- Further Topics and Appendix: reading for familiarity

- Detailed examples and advanced topics are in Further Topics, after the Conclusions
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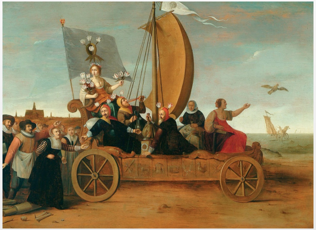
Three Historical Lessons



Hendrik Gerritsz Pot, *Flora's Wagon of Fools* (c. 1637), a satire of the tulip trade

Image: Hendrik Gerritsz Pot (c. 1637); public domain; Wikimedia Commons

- **Tulips, 1636–1637** Garber (1990)
    - Contracts on rare bulbs traded at very high prices until the market stopped in February 1637
    - Part of the path fits the fundamentals of a new luxury good
    - Lesson 1: calling something a bubble requires a model of the fundamental value
- **South Sea and Mississippi, 1720** Kindleberger & Aliber (2005)
    - Shares rose several-fold within months, then collapsed
    - Ingredients: a financial innovation, easy credit, a story about future profits
    - Lesson 2: bubbles come with credit expansion Kindleberger & Aliber (2005) and with high turnover Scheinkman & Xiong (2003)
- **1929** Kindleberger & Aliber (2005)
    - A boom partly financed with borrowed money ended in the October crash
    - Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic
    - Lesson 3: we need tests that work on a single time series, in real time, before the panic

▸ Further Topics: the series tested

The Financial Instability Hypothesis: Minsky

- Minsky classifies borrowers by whether their cash flows can service their debt Minsky (1992)
  - **Hedge** finance: cash flows cover both interest and principal
  - **Speculative** finance: they cover interest, but the principal must be rolled over
  - **Ponzi** finance: they do not even cover interest; debt grows and repayment relies on rising asset prices
- The fragility mechanism
  - A long calm period shifts borrowers towards speculative and Ponzi finance
  - Higher interest rates or falling prices turn speculative borrowers into Ponzi borrowers
  - Forced asset sales push prices down and drive other borrowers towards insolvency
- The link with this chapter
  - The credit that finances the boom: the second case study
  - Accelerating prices: the explosiveness tests

Fundamental Value and the Rational Bubble

- **Question**: which part of the price cannot be justified by dividends, and what can we test about it?
- No-arbitrage pricing: $P_t = \mathbb{E}_t[M_{t+1}(P_{t+1} + D_{t+1})]$
  - P_t: price at the end of period t; D_{t+1}: dividend paid in period $t+1$
  - M_{t+1}: stochastic discount factor; $\mathbb{E}_t$: expectation under the physical measure, given the information at t
- **Simplifying assumption**: $M_{t+1} \equiv 1/(1+r)$, so $P_t = \mathbb{E}_t[P_{t+1} + D_{t+1}]/(1+r)$
  - A constant risk premium does not make M_{t+1} constant
  - In general, the bubble satisfies $B_t = \mathbb{E}_t[M_{t+1} B_{t+1}]$
- Solving forward gives the **fundamental value** $F_t = \sum_{i \geq 1} \mathbb{E}_t[D_{t+i}]/(1+r)^i$
  - The present value of expected dividends
  - Gordon case, dividends growing at $g < r$: $F_t = D_{t+1}/(r-g)$
- But $P_t = F_t + B_t$ also solves the pricing equation, for any B_t with $\mathbb{E}_t[B_{t+1}] = (1+r)\,B_t$
  - B_t is the **rational bubble**: investors pay for it because they expect to resell at a higher price Blanchard & Watson (1982)

Why a Rational Bubble Must Explode

- From $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$: $\mathbb{E}_t[B_{t+h}] = (1+r)^h B_t$ for any horizon of h periods
  - An **explosive conditional mean**
  - Individual paths can crash
  - The bubble grows faster than dividends ($r > g$), so its share of the price rises over time
- Rational bubbles are positive and exist from the first trade Diba & Grossman (1988)
  - A negative bubble would eventually push the expected price below zero
  - A bubble that is zero today must stay zero
- A bubble is the failure of the **transversality condition** $\lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$
  - The price contains a value never paid out as dividends (derivation in the ▸ Appendix)
- **Testable implication**
  - If dividends are at most I(1), then without a bubble the price and the price-dividend ratio are **not explosive**

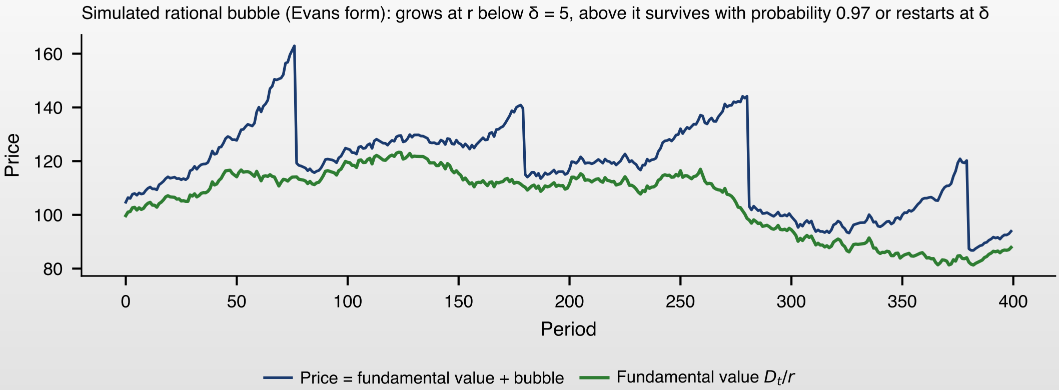
Bubbles That Burst: Blanchard and Watson

- Each period the bubble survives with probability π and collapses with probability $1-\pi$ Blanchard & Watson (1982)
  - $B_{t+1} = \dfrac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π, $B_{t+1} = \varepsilon_{t+1}$ otherwise
  - $\pi \in (0,1)$: survival probability; ε_{t+1}: a shock with mean zero given the information at t
- Expected growth is still r
  - While it survives the bubble grows faster, at $(1+r)/\pi - 1$, to pay for the crash risk
- Expected lifetime: $1/(1-\pi)$ periods, a geometric waiting time

The Evans Form: Bubbles That Restart

- The problem: after a Blanchard–Watson collapse, $B = \varepsilon$ can be negative, which Diba & Grossman (1988) rule out
- Evans form: the bubble starts and restarts from a value $\delta > 0$ Evans (1991)
  - Survival (probability π): $B_{t+1} = \left[\delta + \frac{1+r}{\pi}\left(B_t - \frac{\delta}{1+r}\right)\right] u_{t+1}$
  - Collapse (probability $1-\pi$): $B_{t+1} = \delta\, u_{t+1}$
  - u_{t+1}: positive noise with mean 1, drawn independently of survival
- The bubble condition still holds: $\mathbb{E}_t[B_{t+1}] = \pi\left[\delta + \frac{1+r}{\pi} B_t - \frac{\delta}{\pi}\right] + (1-\pi)\delta = (1+r)B_t$
- Small-bubble branch: if $B_t \leq \alpha$, then $B_{t+1} = (1+r)B_t u_{t+1}$
  - The two branches above apply only for $B_t > \alpha$
  - Without the threshold, the survival branch turns negative for $B_t < \delta(1-\pi)/(1+r)$
  - With $0 < \delta < (1+r)\alpha$ the bubble stays positive; the simulation uses $\alpha = \delta$

A Simulated Rational Bubble (1/2)



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A Simulated Rational Bubble (2/2)

- Simulation
  - Random-walk dividends D_t
  - Discount rate $r = 1\%$, survival probability $\pi = 0.97$
  - Evans form with $\delta = \alpha = 5$
  - Fundamental value: $\mathbb{E}_t[D_{t+i}] = D_t$, so $F_t = \sum_{i \geq 1} D_t/(1+r)^i = D_t/r$
  - While alive and large, the bubble grows by $(1+r)/\pi - 1 \approx 4.1\%$ per period
- The chart: price $P_t = F_t + B_t$ (blue) and fundamental value F_t (green), over 400 periods
- The bubble grew large and burst 4 times
  - At its largest the price was 1.46 times the fundamental value
- Between episodes the price tracks fundamentals
  - A full-sample test sees mostly "normal" data: the tests of this chapter use **windows**

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Bubbles Beyond Rational Expectations

Robert J. Shiller, Nobel Prize 2013

Photo: Presidential Office Building, Taiwan (2017); CC BY 2.0; Wikimedia Commons

- **Feedback and narratives** Shiller (2015)
  - Rising prices attract new buyers
  - “New era” stories spread by word of mouth
- **Disagreement and short-sale limits** Scheinkman & Xiong (2003)
  - The price includes the option to resell to a more optimistic buyer
- **Riding the bubble** Abreu & Brunnermeier (2003), Brunnermeier & Nagel (2004)
  - Informed traders stay in as long as others do, since no one can burst it alone
- Common signature of all three
  - Prices grow faster than fundamentals for a while: the explosive pattern we will test

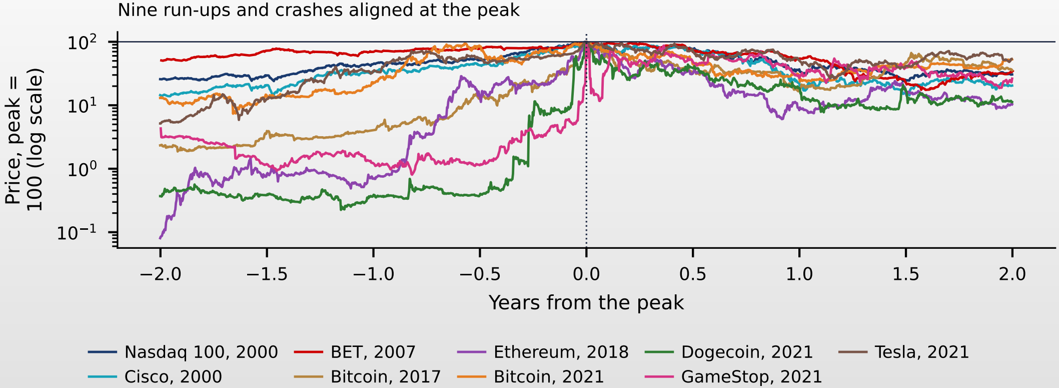
Nine Episodes: Run-Up and Crash

- **Question**: are very fast run-ups followed by crashes, and how often?

Episode	**Peak**	**2-year run-up**	**Max. fall**	**Trough**	**Back at peak**
Nasdaq 100, 2000	27 Mar 2000	+288%	−83%	Oct 2002	Nov 2015
Cisco, 2000	27 Mar 2000	+592%	−89%	Oct 2002	Aug 2021
BET, 2007	24 Jul 2007	+97%	−83%	Feb 2009	Mar 2021
Bitcoin, 2017	16 Dec 2017	+4 186%	−83%	Dec 2018	Nov 2020
Ethereum, 2018	13 Jan 2018	+124 004%	−94%	Dec 2018	Feb 2021
Bitcoin, 2021	8 Nov 2021	+667%	−77%	Nov 2022	Mar 2024
Dogecoin, 2021	7 May 2021	+26 897%	−92%	Jun 2022	not yet
GameStop, 2021	27 Jan 2021	+2 218%	−88%	Apr 2024	not yet
Tesla, 2021	4 Nov 2021	+1 837%	−74%	Jan 2023	Dec 2024

- Peak: highest daily close in the episode
- Run-up: simple percentage change of the price over the two years before the peak
  - $P_{\mathbf{peak}}/P_{\mathbf{peak}-\mathbf{2y}} - 1$
- Maximum fall: to the lowest close within four years
- BET 2007: the only episode without a spectacular run-up in the last two years
  - Yet with one of the deepest falls

Run-Ups and Crashes Aligned at the Peak (1/2)



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Run-Ups and Crashes Aligned at the Peak (2/2)

- Daily prices two years before and after each peak, peak = 100, log scale
- Two-year run-ups differ by orders of magnitude, the falls much less (74–94%)
- Common shape: an accelerating rise, a sharp top, a deep fall
    - Convex on the log scale: the growth rate itself rises, the **explosive** behaviour the tests look for
- Selection bias: we picked these episodes because they crashed
    - Many run-ups were not followed by crashes

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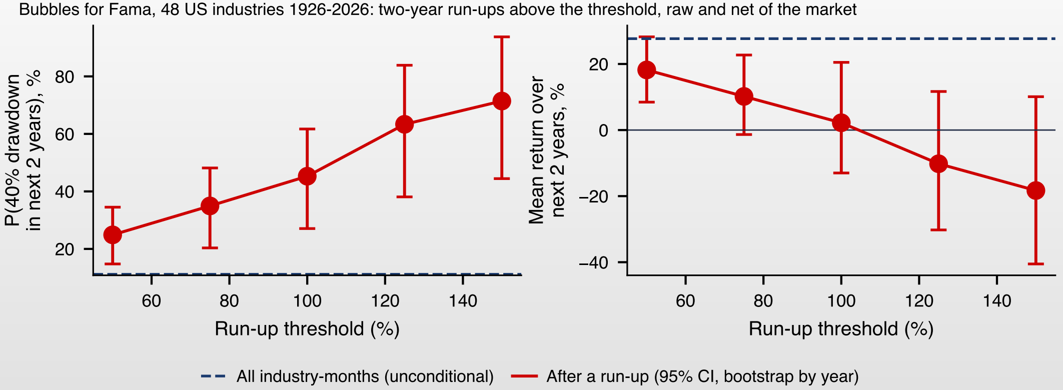
Further Topics: the dot-com bubble

Bubbles for Fama: The Design

- Fama's objection: a "bubble" is only a bubble if its end is predictable Fama (2014)
  - Otherwise it may be a rational reaction to news
- Greenwood, Shleifer and You answer with **all** US industries since 1926 Greenwood, Shleifer & You (2019)
  - Not with famous episodes only
  - Do sharp run-ups predict crashes, or low returns?
- **Run-up**: value-weighted industry return over the past 2 years above a threshold (e.g. 100%)
  - Each firm weighted by its market capitalisation
  - Both conditions at once: the raw return **and** the return net of the market exceed the threshold
  - And a raw return of at least 50% over the past 5 years (from 1931), so that recoveries from slumps are not counted
  - First month of the run-up; a new episode in the same industry only two years later
- **Crash**
  - A drawdown of at least 40% from a previous high within the next 2 years
  - Each crash frequency comes with a 95% CI (Confidence Interval)
- The sample: the first 48 of the 49 industries of the Kenneth French Data Library (without "Other")
  - Months with at least ten firms: 41 721 industry-months, 1926–2026

What Do Sharp Run-Ups Predict? (1/2)



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What Do Sharp Run-Ups Predict? (2/2)

- Crash frequency above threshold c: $\hat{p}(c) = n_{\mathbf{crash}}(c)/n(c)$
  - The share of run-ups above c followed by a 40% crash
  - The chart: left, $\hat{p}(c)$ with its 95% CI at the thresholds 50, 75, 100, 125 and 150%; right, the mean 2-year return; dashed line: all industry-months
- After a run-up above 100%, a 40% crash followed in 45% of cases
  - 53 episodes in 27 industries
  - 95% CI $[27\%, 62\%]$
  - Against 11% for a random industry-month
  - Above 150%: 71% ($n = 21$)
- CI: bootstrap over calendar years, because run-ups cluster in time
  - E.g. 1928–1929, 1967–1968, 1980
  - Effective sample size: about 31 independent observations (▸ Appendix)
- The mean return over the next 2 years is not significantly different from zero
  - 2% after a 100% run-up, CI $[-13\%, 20\%]$
- Run-ups raise the **risk** of a crash, not the predictable loss Greenwood, Shleifer & You (2019)
  - The result does not establish the profitability of a market-timing strategy
- What the result supports: association
  - a conditional frequency: the run-up precedes the crash, with no mechanism identified

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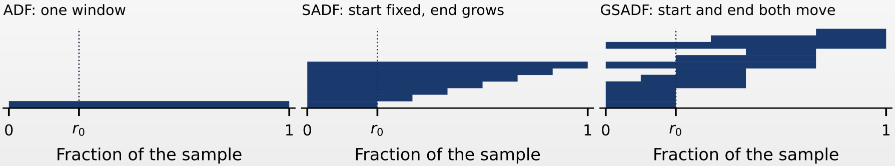
Right-Tailed ADF (1/2): Regression and Hypothesis

- **Question**: how do we show, on a single price series, that there was an explosive phase and when it started?
- Regression on the log price y_t: $\Delta y_t = a + b\, y_{t-1} + \varepsilon_t$
  - $\Delta y_t = y_t - y_{t-1} = \ln(P_t / P_{t-1})$: the log return; ε_t: shock; here no lagged Δy
  - $\mathrm{ADF} = \hat{b}/\mathrm{SE}(\hat{b})$: the Augmented Dickey–Fuller (ADF) t-statistic; $\hat{b}$: the OLS (Ordinary Least Squares) estimate; SE: its Standard Error
  - $b < 0$: mean reversion; $b = 0$: random walk (unit root); $b > 0$: each rise feeds the next, an explosive path
- Bubble test: $H_0 : b = 0$ (unit root) vs $H_1 : b > 0$ (explosive root), the **right tail** Phillips, Wu & Yu (2011)
  - The left-tail Dickey–Fuller table does not apply

Right-Tailed ADF (2/2): Collapsing Bubbles

- Evans: a bubble that collapses periodically leaves the **full-sample** test with low power Evans (1991)
  - The whole series looks stationary or like a random walk
  - Collapses give the full-sample slope a negative sign (▸ Appendix)
- Solution: run the ADF regression on many **windows** and take the largest statistic
  - SADF (Sup Augmented Dickey–Fuller) Phillips, Wu & Yu (2011) and GSADF (Generalised Sup Augmented Dickey–Fuller) Phillips, Shi & Yu (2015a) differ in the windows they search

Three Ways to Choose Windows (1/2)


ADF: one window
SADF: start fixed, end grows
GSADF: start and end both move
0
r_0
1
Fraction of the sample
0
r_0
1
Fraction of the sample
0
r_0
1
Fraction of the sample

- $\mathrm{ADF}_{r_1}^{r_2}$: the ADF statistic on the window from fraction r_1 to fraction r_2 of the sample
  - Each bar in the chart is one ADF regression
  - r_0: the minimum window, as a fraction of the T changes Δy_t; sup: the largest value over the windows allowed
- Minimum window rule: $r_0 = 0.01 + 1.8/\sqrt{T}$ Phillips, Shi & Yu (2015a)
  - E.g. $T = 440$ months: $r_0 = 0.096$, i.e. 42 months
  - Too short a window: an imprecise ADF and more false alarms; too long: short bubbles are missed
  - $w_0 = r_0 T = 0.01T + 1.8\sqrt{T}$: the minimum share r_0 falls towards 1%, but the minimum number of observations grows
  - An episode of fixed length can become too short for the minimum window of a very long sample

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Three Ways to Choose Windows (2/2)

- SADF Phillips, Wu & Yu (2011)
    - $\mathrm{SADF}(r_0) = \sup_{r_2 \in [r_0, 1]} \mathrm{ADF}_0^{r_2}$
    - Windows start at the first observation and end anywhere after r_0
- GSADF Phillips, Shi & Yu (2015a)
    - $\mathrm{GSADF}(r_0) = \sup_{r_2 \in [r_0, 1]} \ \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$
    - Windows start and end anywhere, with length at least r_0
- BSADF (Backward Sup Augmented Dickey–Fuller): backward SADF at the end point r_2 Phillips, Shi & Yu (2015a)
    - $\mathrm{BSADF}_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}, \ \ \mathrm{GSADF} = \sup_{r_2} \mathrm{BSADF}_{r_2}$
    - BSADF_{r_2} uses only data up to r_2: it can be computed **in real time**
- A larger search needs a higher bar: the critical value is simulated for each T, r_0 and lag order

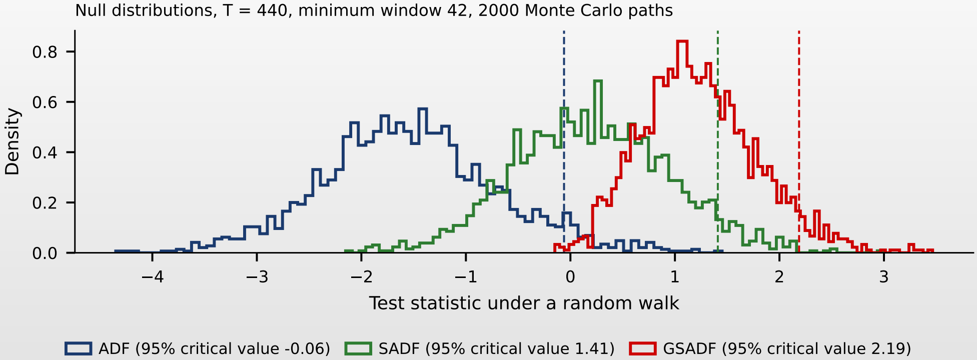
Date-Stamping an Episode (1/2): the PSY Rules

- **Origination**, in the PSY (Phillips, Shi and Yu) procedure Phillips, Shi & Yu (2015a)
  - The first date at which BSADF_{r_2} exceeds its critical value cv_{r_2}
  - cv_{r_2}: the 95% quantile of BSADF_{r_2} under the null, date by date
- **Termination** (PSY, eq. 8) Phillips, Shi & Yu (2015a)
  - The first date after $\hat{r}_e + \delta \ln(T)/T$ at which BSADF falls below cv_{r_2}
  - $\hat{r}_e$: the estimated start, as a fraction of the sample; T: number of changes Δy_t
  - $\delta \ln(T)/T$: a minimum duration, with δ chosen for the data frequency
  - Why $\ln T$: an isolated exceedance often occurs by chance; a duration that grows slowly with T removes short alarms without losing long bubbles
- PWY (Phillips, Wu and Yu) date-stamping uses the recursive $\mathrm{ADF}_0^{r_2}$ instead Phillips, Wu & Yu (2011)
  - It may miss a second bubble after a collapse

Date-Stamping an Episode (2/2): Minimum Duration, Real Time and Chains

- Minimum-duration rule of the R package exuber Vasilopoulos, Pavlidis & Martínez-García (2022)
  - Runs of consecutive exceedances shorter than L periods are dropped
  - $L = \lceil \ln T \rceil$, i.e. $\delta = 1$; $\lceil \cdot \rceil$: rounded up; $T = 440$ months gives $L = 7$
- Start: the first exceedance; end: the last exceedance
  - The PSY rule (eq. 8) would turn a single exceedance into an episode of L periods
- Real time: the episode is **confirmed** only after L consecutive exceedances
  - $L - 1$ periods after its start, which is dated in hindsight
- **Chain**: kept runs separated by a short gap
  - Monthly data: the next run starts at most 93 days after the previous one ends (at most two months without an exceedance)

Null Distributions and Critical Values (1/2)



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Null Distributions and Critical Values (2/2)

- Random walk with a small drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$, 2 000 paths of $T = 440$
- 95% critical values Phillips, Shi & Yu (2015a)
  - A single ADF −0.06 (negative)
  - SADF 1.41, GSADF 2.19
- The supremum over about 80 000 windows shifts the GSADF distribution to the right
- Asymptotics (▸ Appendix)
  - The null limit is pivotal
  - Against mildly explosive roots the tests are consistent
- Date-stamping keeps a 5% false-alarm chance at every date Phillips, Shi & Yu (2015b)

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Changing Volatility: Wild Bootstrap

- Monte Carlo critical values simulate $y_t = T^{-1} + y_{t-1} + \varepsilon_t$, $\varepsilon_t \sim N(0,1)$
  - Shocks with constant variance
  - Real prices show volatility clustering and shifts in volatility
  - A burst of volatility can mimic explosiveness and inflate the rejection rate Harvey et al. (2016)
- **Wild bootstrap** Phillips & Shi (2020)
  - $\Delta y_t^* = w_t(\Delta y_t - \overline{\Delta y})$, with $w_t = \pm 1$, each with probability 1/2
  - $y_t^* = y_0 + \sum_{s \leq t} \Delta y_s^*$; then recompute BSADF on y^*
  - The bootstrap data have a unit root by construction (no explosive root) and inherit the time pattern of $\mathrm{Var}(\Delta y_t)$ Harvey et al. (2016)
  - Demeaning removes the drift, as in the null model
- Retrospective and real-time versions
  - Centring with the full-sample mean and resampling the whole path give **retrospective** critical values
  - In real time, recompute them with data up to each date

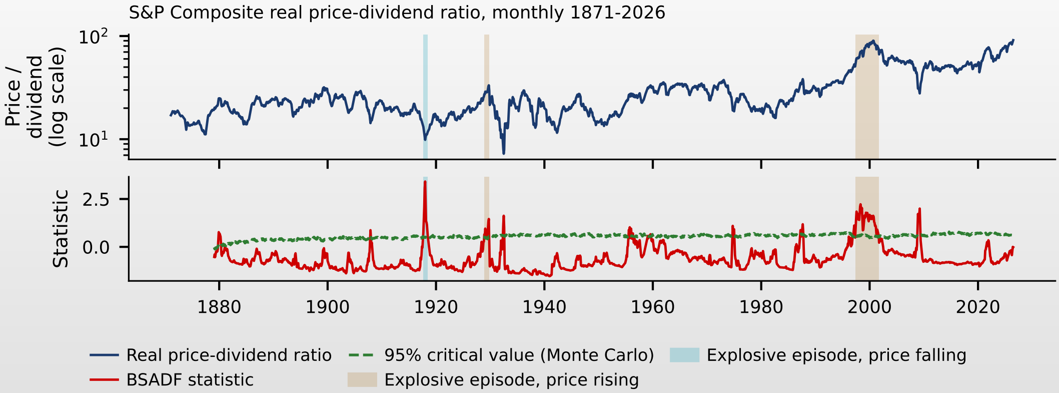
Further Topics: GSADF size, simulation

Serial Dependence: A One-Lag Check

Series	$\hat{\rho}_1$	**Ljung–Box p**	**GSADF,** $k = 0$	**GSADF,** $k = 1$	**95% critical value,** $k = 1$
Nasdaq 100, monthly	0.05	0.58	2.57	3.08	2.40
Bitcoin, weekly	0.07	0.35	3.85	3.60	2.37
BET, monthly	0.18	0.008	2.90	2.29	2.39

- The chapter's ADF regressions have no lags of Δy; autocorrelated returns can distort their size
  - $\hat{\rho}_1$: first-order autocorrelation of returns
  - Ljung–Box p: the test with 12 lags, not corrected for heteroskedasticity
  - k: number of lags of Δy in the ADF regression
  - The Monte Carlo critical value is recomputed for the one-lag regression (2 000 paths)
- Nasdaq 100 and Bitcoin: the conclusion is the same with one lag
- BET: monthly returns are autocorrelated, and with one lag GSADF falls below its critical value ($p = 0.063$)
- The wild bootstrap keeps the time pattern of the variance, not the autocorrelation: independent signs destroy it
  - Nor does it cover infinite variance

S&P Composite (Shiller) Price-Dividend Ratio, 1871–2026 (1/2)



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S&P Composite (Shiller) Price-Dividend Ratio, 1871–2026 (2/2)

- Real price over real dividend, monthly, Jan 1871–Jun 2026
  - $T = 1\,865$, minimum window 96 months
  - In the chart legend, "price rising/falling" refers to the plotted series, here the ratio
- GSADF $=$ 3.41 vs 95% critical value 2.39 (99%: 2.88)
  - The full-sample ADF (-1.11) sees nothing
- Episodes: Aug 1917–Jun 1918; Nov 1928–Sep 1929; May 1997–Aug 2001
  - 1917–1918: an accelerating **fall** of the ratio, explosive but not a bubble: always report the direction
- Dot-com: peak 90.2 in Aug 2000, long-run mean 29.8
  - The sample maximum, 91.2 in May 2026, lies in no episode: a high ratio is not necessarily explosive
- A ratio can explode through its denominator: check both price and dividend
- What the result supports: description
  - the test dates explosive episodes; it does not separate them from explosive fundamentals

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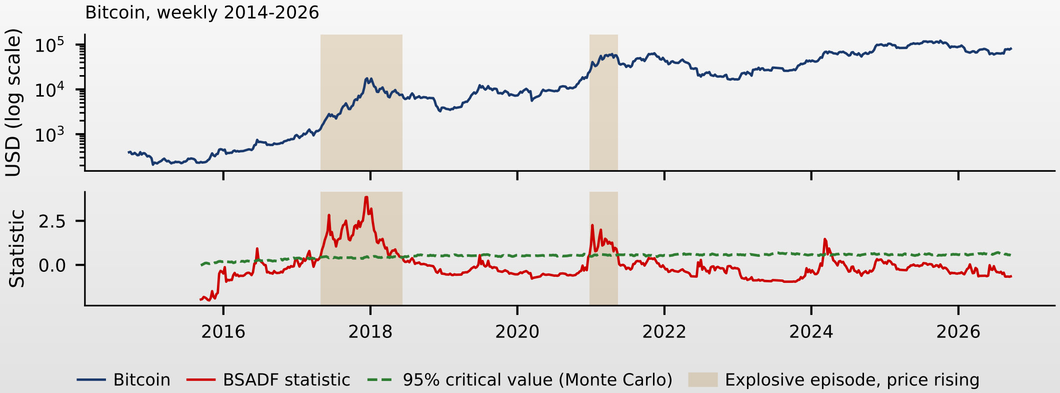
Further Topics: Nasdaq 100, GSADF vs PWY

Bitcoin: Explosive Episodes (1/2)



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Bitcoin: Explosive Episodes (2/2)

- Weekly log closes in US dollars (USD) 2014–2026, $T = 626$, minimum window 51 weeks
- GSADF $= 3.85$: rejected with either critical value
  - 95% critical value: Monte Carlo 2.27, wild bootstrap 2.82
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
  - Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- The 2017 episode starts 8 months before the peak of 16 Dec 2017 and ends 6 months after it
- The 2024–2025 rise to new highs is **not** flagged: steady growth is not explosive acceleration
- What the result supports: description

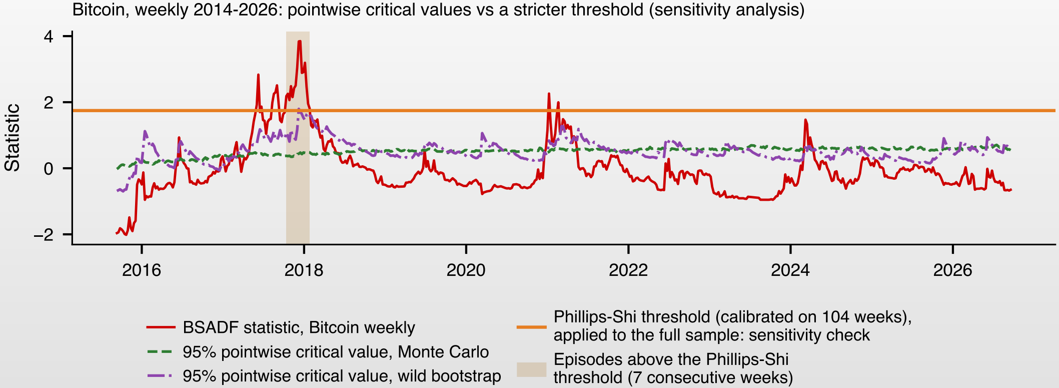
MFM_ch17_bitcoin

Date-Stamping Is a Multiple Test: Bitcoin (1/2)

- BSADF vs its critical value at each of 626 weeks: hundreds of dependent tests
- Family-wise error rate (FWER)
  - The probability of at least one false alarm over the whole monitoring window
  - 2 000 random walks with the Bitcoin T and $w_{\mathbf{0}}$: a crossing in 95%, an "episode" in 59% of paths
- Phillips and Shi: bootstrap the **maximum** of BSADF Phillips & Shi (2020)
  - Over the last τ_b monitoring dates; here $\tau_b = 104$ weeks
  - Phillips–Shi threshold: 1.75, against Monte Carlo pointwise values of at most 0.71
- Episodes above the Phillips–Shi threshold applied to the whole history: Oct 2017–Jan 2018
  - The 2020–2021 episode found with Monte Carlo pointwise values does not survive
- A sensitivity check, not proved FWER control (design in the ▸ Appendix)
  - The threshold is calibrated on 104 weeks; applied to all 626, under the null it is crossed in 19% of paths
  - A confirmed episode (7 consecutive crossings) occurs in 2% of paths

MFM_ch17_inference

Date-Stamping Is a Multiple Test: Bitcoin (2/2)



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Real Time vs Hindsight

Episode	Start / Peak	Confirmed	Lead (months)	Lead from conf.	Rise after conf.	Fall after peak
Nasdaq 100, monthly	Jun 1995 / Mar 2000	Dec 1995	57	51	+716%	−76%
BET, monthly	Nov 2003 / Jul 2007	Apr 2004	44	39	+299%	−83%
Bitcoin 2017, weekly	Apr 2017 / Dec 2017	Jun 2017	8	6	+590%	−83%
Bitcoin 2021, weekly	Dec 2020 / Apr 2021	Feb 2021	4	2	+66%	−75%
GameStop, daily	13 Jan 2021 / 27 Jan 2021	22 Jan 2021	0.5	0.2	+435%	−88%

- Columns (caveats in the ▸ Appendix)
  - Start: first exceedance of the chain of BSADF runs that leads to the peak (hindsight); confirmed: after L consecutive exceedances (real time)
  - Rise: from confirmation to the peak; fall: to the lowest close within 2 years (GameStop: within 1 year)
- Even at confirmation the tests **warn early**, but they do not time the crash Phillips, Shi & Yu (2015b)
  - Selling at confirmation would have missed most of the rise
- What the result supports: description
  - five episodes that ended in crashes: the lead is not a hit rate of the alarms

Reading the Real-Time Table

- Largest gap inside a chain
  - 93 days for monthly data
  - 28 days for weekly data
  - 10 days for daily GameStop
- Nasdaq 100: the chain joins 3 runs, from Jun 1995
  - The strict rule keeps only the run containing the peak
  - Strict rule: start Sep 1998, confirmed Mar 1999, leads 18 and 12 months instead of 57 and 51
  - For BET, Bitcoin and GameStop the two rules give the same start
- Two falls for the same dot-com crash
  - This table: −76%, the lowest daily close within 2 years of the peak
  - Nine Episodes table: −83% within 4 years; the trough of Oct 2002 came 30 months after the peak, outside the 2-year window
- Bitcoin 2021: the peak here is the local high of 13 Apr 2021, not the all-time high of 8 Nov 2021
  - The explosive chain ended in May 2021; the later high came after a deep fall and a slower recovery that BSADF did not flag

Crashes as Critical Points

Didier Sornette, ETH Zurich

Photo: Didier Sornette (2012); CC BY-SA 3.0 (Germany); Wikimedia Commons

- **Question**: can the shape of the run-up tell when the bubble ends?
- Johansen, Ledoit and Sornette (JLS) Johansen, Ledoit & Sornette (2000)
  - Imitation among traders makes the **hazard rate** of a crash rise as a critical time t_c approaches
  - Hazard rate: the crash probability per unit time, given no crash yet
- Rational investors stay in only if the price grows fast enough
  - The extra growth pays for the rising crash risk
- Result: **faster-than-exponential** growth
  - Decorated by oscillations that speed up near t_c
- t_c is the most probable end of the bubble, not a certain crash date Sornette (2017)

The LPPLS Equation

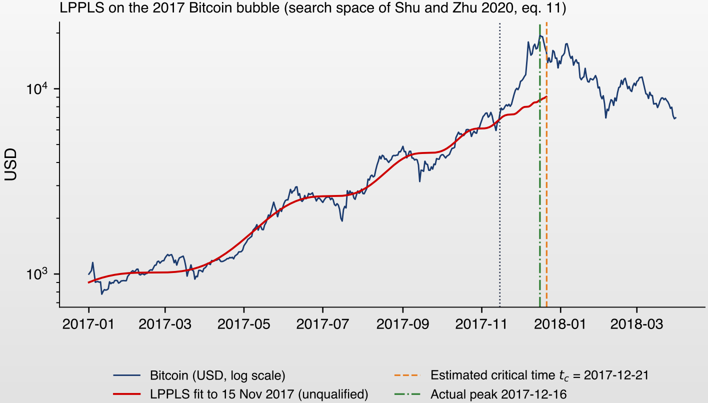
- $\ln p(t) = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi), \quad t < t_c$
  - In words: the log price is a power-law trend towards the critical time plus oscillations whose frequency rises as t_c approaches; $p(t)$: price at time t; t_c: critical time
- A: log price at t_c
- $B < 0$: the super-exponential trend for a rising bubble
- $0 < m < 1$: power-law exponent
  - The growth rate of the power-law trend, $-Bm(t_c - t)^{m-1}$, rises without bound as $t \to t_c$
- C, ω, ϕ: amplitude, angular log-frequency and phase of the log-periodic oscillations
- Seven parameters: A, B, C, ϕ and the nonlinear t_c, m, ω
- LPPL: the log-periodic power law pattern
  - LPPLS: the model with the finite-time singularity at t_c

Further Topics: LPPLS calibration

Qualified Fits: Conditions on the Parameters and the Residuals

- A fit on the estimation window $[t_1, t_2]$ is **qualified**
  - If it passes all conditions of Shu & Zhu (2020a, eq. 12)
  - And $B < 0$ for a rising bubble
  - $m \in [0.01, 0.99]$, $\omega \in [2, 25]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/5]$
  - Damping $\frac{m|B|}{\omega|C|} \geq 1$ (already imposed in the search)
  - Number of oscillations (half-periods) $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$, so the oscillations are not noise
  - Largest relative error of the fitted price $\max_t |\hat{p}_t - p_t|/p_t \leq 0.15$
- Lomb periodogram of the detrended residual $(t_c - t)^{-m}(\ln p_t - A - B(t_c - t)^m)$
  - A heuristic filter with nominal level 0.10: the trend is fitted on the same data, so the p-value is not calibrated
  - Lomb periodogram: spectral power for unevenly spaced points, here the values of $\ln(t_c - t)$
- The residual $\ln \hat{p}_t - \ln p_t$ must be stationary
  - Dickey–Fuller and Phillips–Perron reject a unit root at the nominal level 0.10, again a heuristic filter
  - Phillips–Perron: a Dickey–Fuller-type test with a nonparametric correction for serial correlation

LPPLS on Bitcoin, 2017 (1/2)



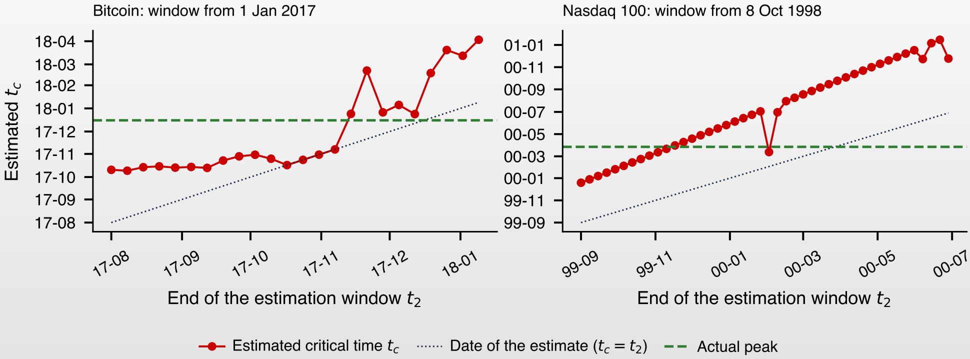
MFM_ch17_lppls ▶ **Interactive**

LPPLS on Bitcoin, 2017 (2/2)

- Fit on daily log prices, 1 January to 15 November 2017
  - The path after 15 November is out of sample
- The fit reproduces the 2017 trend and oscillations, but its largest relative error exceeds 0.15
  - $m = 0.95$, $\omega = 10.54$, $B = -2.62$
  - Damping $D = 1.00$, at the bound imposed in the search; the exact condition $m|B| \geq |C|\sqrt{m^2 + \omega^2}$ fails (ratio 0.996)
- Estimated t_c: 21 Dec 2017; actual peak: 16 Dec 2017
- The fit does **not** pass the filter, even though it looks convincing

MFM_ch17_lppls

How Stable Is the Critical Time? (1/2)



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How Stable Is the Critical Time? (2/2)

- A good in-sample fit is not a forecast
  - Each point: one fit on a window with a fixed start, ending on the date on the horizontal axis (every 7 days)
- Bitcoin: before the peak (20 fits) t_c ranged from Oct 2017 to Feb 2018
  - Mean absolute error 48 days; within 30 days of the peak: 20% of the fits
- Nasdaq 100: before the peak (30 fits) t_c ranged from Dec 1999 to Sep 2000
  - Mean absolute error 71 days; $t_c \geq t_2$ is a constraint of the search, not a forecast
- Qualified fits before the peak: 0% (Bitcoin), 0% (Nasdaq 100)
  - One window is not enough: the confidence indicator pools many windows

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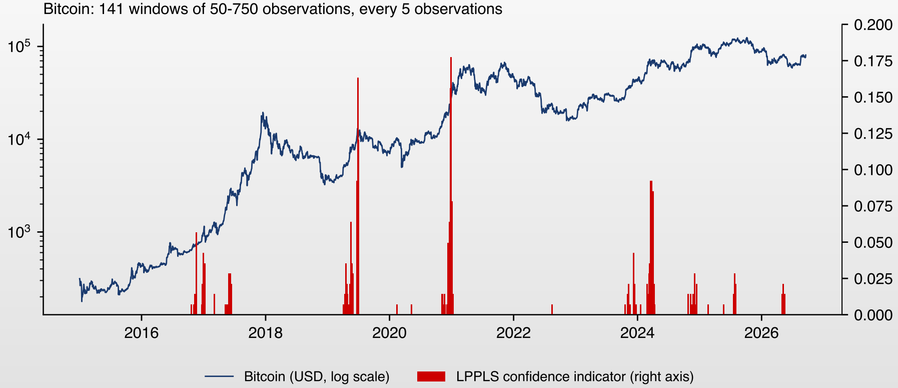
The LPPLS Confidence Indicator: Definition and Evaluation

- At each date t_2, fit many windows $[t_2 - \ell, t_2]$ and report the **share of qualified fits** Demirer et al. (2019)
  - $I_{t_2} = \frac{1}{N} \sum_{\ell} 1\{\text{the fit on } [t_2 - \ell, t_2] \text{ is qualified}\}$, between 0 and 1
  - Grid of Shu and Zhu: $N = 141$ windows, ℓ from 750 down to 50 observations in steps of 5 Shu & Zhu (2020a)
  - t_2 moves every 5 observations (calendar days for Bitcoin); data up to t_2 only
- A high share means the bubble signature is robust to the choice of window
- The birth of a bubble is easier to diagnose than its burst Demos & Sornette (2017)
- Evaluation: is a signal ($I_{t_2} > 0$) followed by a fall of at least 20% within 90 days
  - More often than no signal?
  - Fall measured from the price on the signal date
  - Only dates whose next 90 calendar days are fully observed
  - The 20% threshold and the 90-day horizon are fixed in advance; the AI section checks a grid of 4 horizons × 3 thresholds

Bitcoin: LPPLS Confidence Indicator



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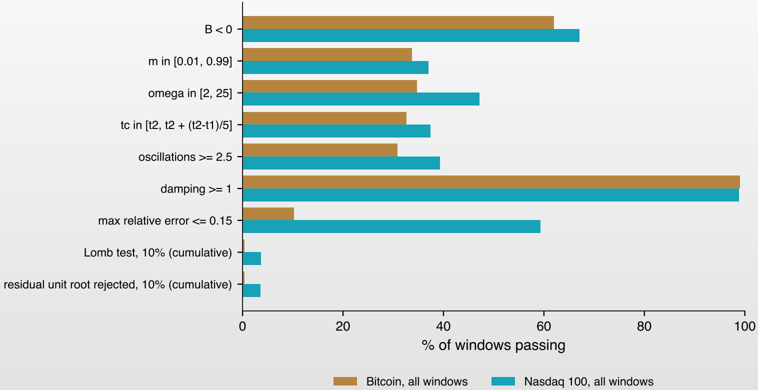
LPPLS: Qualification Filters



MFM_ch17_lppls_confidence ▶ **Interactive**

The Confidence Indicator and Its Filters: Results

- Price panel: Bitcoin price and the share of the 141 windows with a qualified fit
  - The share is shown as bars on the right axis, every 5 days, 2016–2026
  - Filter panel: share of all windows passing each condition, Bitcoin and, for comparison, the Nasdaq 100
  - Cumulative rows: the Lomb and residual tests count only windows that pass every condition above them
- Positive on 14% of the 2017 dates, 3% in 2021, 12% in other years
  - Maximum 0.18 in Dec 2020
- A 20% fall within 90 days followed 22% of signals, 37% of no-signal dates
  - Bitcoin: 709 dates; Nasdaq 100, 1997–2026: 4% vs 11% (1483 dates)
- The raw difference is not yet evidence (test on the next slide)
  - Signals come in runs and the 90-day outcome windows overlap

MFM_ch17_lppls_confidence

Is the Indicator Better than Chance? A Test (1/2)

- H_0: $P(\text{fall} \geq 20\% \text{ within 90 days} \mid \text{signal}) = P(\cdot \mid \text{no signal})$
  - Statistic $\Delta = \hat{p}_{\mathbf{on}} - \hat{p}_{\mathbf{off}}$: share of signal dates followed by such a fall minus the share of no-signal dates
- Overlap: dates 5 days apart share up to 85 of their 90 outcome days
  - The two-sample z-test assumes independent dates and so understates the standard error
- Dependence-robust inference
  - Circular block bootstrap of the (signal, outcome) pairs: blocks of 90 days, 9 999 replications
  - Linear probability regression of the outcome on the signal, with Newey–West standard errors with 18 lags (90 days at 5-day spacing)

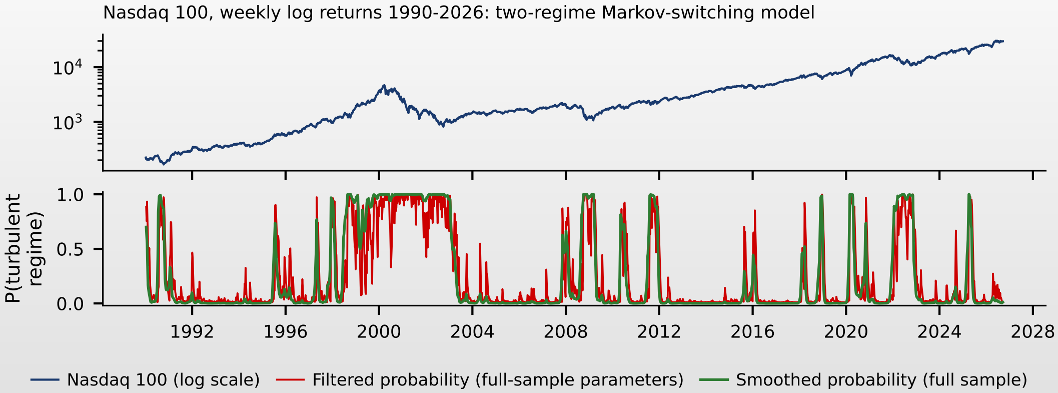
Is the Indicator Better than Chance? A Test (2/2)

- Bitcoin: $\Delta = -15.5$ percentage points, 95% block-bootstrap CI $[-30.0, 1.8]$, which contains zero
  - Block SE 8.1 vs independent-data SE 5.0
  - Blocks of 36 and 72 dates: CI $[-30.8, 2.4]$ and $[-30.2, 1.0]$
- Newey–West regression: two-sided $p = 0.049$ for $H_0 : \Delta = 0$, at the edge of the 5% level
  - The percentile CI and the Newey–West t-test use different standard errors, so they need not agree near the 5% boundary
- Weak evidence, and the point estimate is negative: if anything, a lower crash rate after signals
  - Signals come in steady rises, not just before crashes
  - The negative direction was not stated in advance: the one-sided $p = 0.024$ is only exploratory
- What the result supports: association
  - between the signal and later falls: weak, and the point estimate is negative

The Markov-Switching Model

- **Question**: can run-ups and crashes be described as alternating risk states, without dating a bubble?
- Weekly log return $r_t = \mu_{s_t} + \sigma_{s_t}\varepsilon_t$, $\varepsilon_t \sim N(0,1)$, regime $s_t \in \{1,2\}$ unobserved Hamilton (1989)
  - $r_t = 100\ln(P_t/P_{t-1})$, in %
  - Each regime has its own mean μ and volatility σ: a calm and a turbulent regime
- Regimes follow a Markov chain: $p_{ij} = P(s_t = j \mid s_{t-1} = i)$
  - Expected duration of regime i: $1/(1-p_{ii})$
- Estimation by maximum likelihood with the Hamilton filter
  - The filter updates $P(s_t = j \mid r_1, \dots, r_t)$ week by week
- **Filtered** probability: returns up to t for given parameters
- **Smoothed** probability: the whole sample (hindsight)
  - Our parameters are estimated on the full sample, so our filtered probabilities are only **pseudo real time**; genuine real time needs parameters re-estimated with data up to t

Nasdaq 100: Two Regimes (1/2)



MFM_ch17_markov_switching ▶ **Interactive**

Nasdaq 100: Two Regimes (2/2)

Regime	$\hat{\mu}$ (SE)	$\hat{\sigma}$	$\hat{p}_{ii}$	Dur.
Calm	0.43 (0.06)	2.30	0.987	75.4
Turbulent	−0.32 (0.27)	5.40	0.957	23.5

- Weekly log returns in %, 1990–2026, 1 915 weeks
  - Duration in weeks; calm regime in 76% of the weeks
- Turbulent: volatility more than twice as high, negative (insignificant) mean
- The turbulent spell around the 2000 peak began in Jun 1998 (smoothed probability above 0.5)
  - Filtered probability 0.99 at the end of March 2000, in the first falling week after the peak
  - Weekly close: from 4 691.61 (24 March, the peak) to 4 397.84 (31 March), a log return of −6.5%

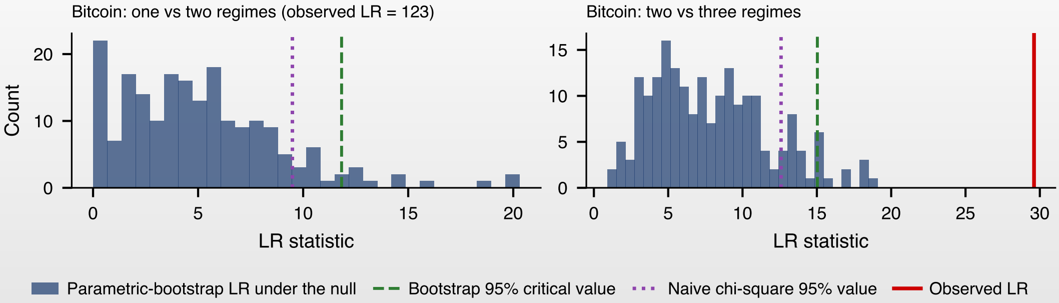
MFM_ch17_markov_switching

Further Topics: Bitcoin regimes

How Many Regimes? A Bootstrap LR Test (1/3)

- H_0: one regime (iid Normal returns) vs H_1: two Markov regimes
  - Statistic $\mathrm{LR} = 2(\hat{\ell}_2 - \hat{\ell}_1)$; $\hat{\ell}_1$, $\hat{\ell}_2$: maximised log-likelihoods with one and two regimes
- Non-standard null distribution (▸ Appendix) Hansen (1992)
  - p_{11}, p_{22} are not identified under H_0, so LR is not χ^2
- **Parametric bootstrap**
  - Simulate from the fitted H_0 model, refit both models, recompute LR; $B = 199$ replications, so the smallest attainable p is $1/(B+1) = 0.005$
  - $p = (1 + \#\{\mathrm{LR}^* \geq \mathrm{LR}\})/(B+1)$; $\#\{\cdot\}$: number of bootstrap values LR^* at least as large as the observed LR

How Many Regimes? A Bootstrap LR Test (2/3)



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How Many Regimes? A Bootstrap LR Test (3/3)

- One vs two regimes: bootstrap $p = 0.005$, the smallest attainable
  - After refitting, 5 (Bitcoin) and 1 (Nasdaq 100) of 199 bootstrap LRs stay negative, of order 10^{-3}
  - Bitcoin: LR $=$ 123.2, 95% critical value 11.8, largest replication 20.3
  - Nasdaq 100: LR $=$ 475.7, 95% critical value 11.2, largest replication 18.1
- Two vs three regimes, Bitcoin: LR $=$ 29.6, above every replication (maximum 19.1), so again $p = 0.005$
  - Bootstrap 95% value 15.0 vs naive χ^2_6 value 12.6
- One regime is rejected for both series
  - A third Bitcoin regime: an estimated transition probability of order 10^{-9}, so a boundary maximum
  - 154 of 199 replications also end on the boundary: the three-regime p is only indicative
  - Caveat: the one-regime null is Normal, so fat tails alone can favour two regimes
- What the result supports: description

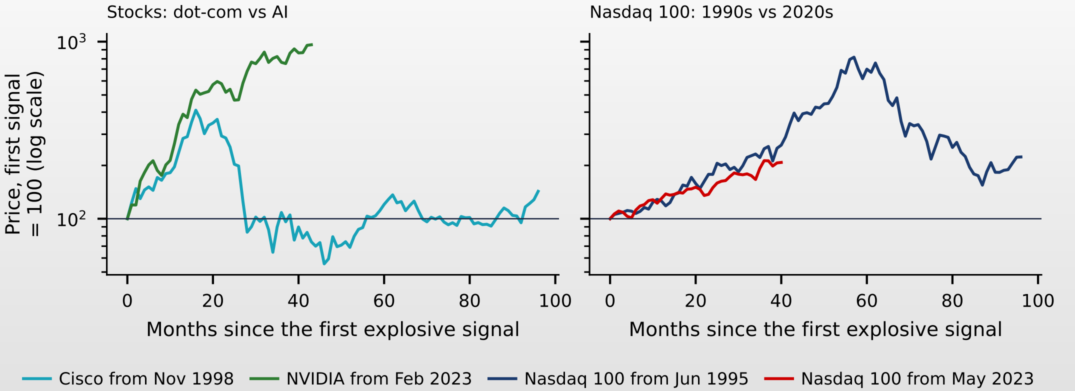
MFM_ch17_inference

Are Crypto-Assets Bubbles?

- **Question**: without a fundamental value, can we still test the price: is it explosive, and when?
- No dividends: the fundamental value of Bitcoin is hard to define
    - One estimate puts it at zero Cheah & Fry (2015)
- Alternative: value from network adoption (Metcalfe's law), with LPPLS bubbles on top Wheatley et al. (2019), Pele & Mazurencu-Marinescu-Pele (2019)
- Crypto-assets increasingly behave like an alternative asset class, not a pure bubble Pele et al. (2023)
- Part of the 2017 Bitcoin rise has been tied to Tether issuance Griffin & Shams (2020)

▸ Further Topics: crypto and tech episodes

Dot-Com vs AI (1/2)



MFM_ch17_crypto_ai ▶ Interactive

Dot-Com vs AI (2/2)

- Monthly closes, rebased to 100 at the first explosive signal of each episode (BSADF, monthly data)
  - First signal: start of the first date-stamped run beginning in 1995 or later (dot-com) or in 2023 or later (AI), not of the run that contains the peak
- Cisco: first signal Nov 1998; NVIDIA: first signal Feb 2023
  - Cisco peaked after 16 months at 4.1 times the signal price
  - NVIDIA, 43 months later: 9.6 times, episode still open
- Nasdaq 100: the AI rise is slower than in the 1990s
  - From Jun 1995: peak after 57 months at 8.2 times; from May 2023: 2.1 times after 40 months

MFM_ch17_crypto_ai

Is There an AI Bubble? What the Tests Can Say

- Monthly BSADF flags NVIDIA and the Nasdaq 100
  - NVIDIA: Mar 2016–Nov 2018; Feb 2020–May 2022; Feb 2023–ongoing; Nasdaq 100: current episode from May 2023
  - Monthly dates ($T = 332$, $w_0 = 36$, $L = 6$ months); the weekly timeline ($T = 1\,443$, $w_0 = 82$, $L = 8$ weeks) shifts the dates by up to a few months, because T, w_0 and L change with the frequency
- The tests cannot separate a bubble from explosive fundamentals Pástor & Veronesi (2009)
  - Fast-growing expected profits of a new technology can also make prices explosive
- What we can say
  - The tests detect explosive price dynamics, not their cause
  - The industry run-up evidence Greenwood, Shleifer & You (2019) motivates watching crash risk, but does not quantify it for these stocks
  - The timing of any crash is unknown
- What the result supports: description

Case Study (1/5): the Research Question

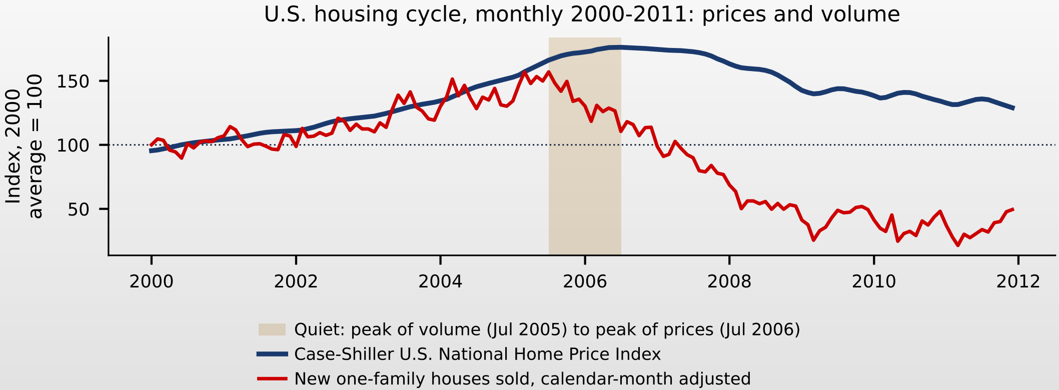
- Paper: DeFusco, Nathanson & Zwick (2022), *Journal of Financial Economics*, 146(1), 205–229
  - Setting: the US housing cycle of 2000–2011, the boom and bust behind the 2008 crisis
- Questions
  - Who buys during a boom?
  - Can speculation explain the joint path of prices, transaction volume and unsold listings?
- Speculation, in the Harrison–Kreps definition the authors adopt
  - Paying more for an asset because one has the right to resell it
- Three phases
  - Boom: prices and volume rise
  - "Quiet": volume falls while prices still rise and unsold listings pile up
  - Bust

▸ Further Topics: data and method

Case Study (2/5): Headline Results

- Figures 2 and 4
  - The correlation peaks at a lag of 24 months
  - Short-term sales explain 39% of the 2000–2005 rise in sales
- Table 2: cities with more speculation had larger booms and larger busts
  - One standard deviation more speculative volume: boom larger by 24.9 points, bust deeper by 7.4 points
- Model (Sections 5–6, Table 6): extrapolating short-horizon buyers amplify the cycle
  - A tax of 5% of the purchase price, paid at sale, cuts the simulated log-price bust from 8.2 to 1.1 log points
- What the result supports: association
  - between speculation and the size of the cycle, across cities; the tax effect comes from the model, not the data

Case Study (3/5): Prices and Volume, 2000–2011



MFM_ch17_case_study ▶ **Interactive**

Case Study (4/5): Prices and Volume, 2000–2011

- FRED (Federal Reserve Economic Data)
    - Case-Shiller U.S. National Home Price Index
    - New one-family houses sold
    - 2000 average = 100; volume adjusted by calendar-month means
- Boom: sales peak at 157 in Jul 2005
    - Prices keep rising to 176 in Jul 2006
- Quiet: 12 months from the volume peak to the price peak
    - Sales already down to 110 at the price peak (unadjusted sales: 16 months)
- Bust: 2011 averages 35 for sales, 133 for prices
    - New-home sales exclude resales, so they cannot show who trades

MFM_ch17_case_study

Further Topics: volume leads prices

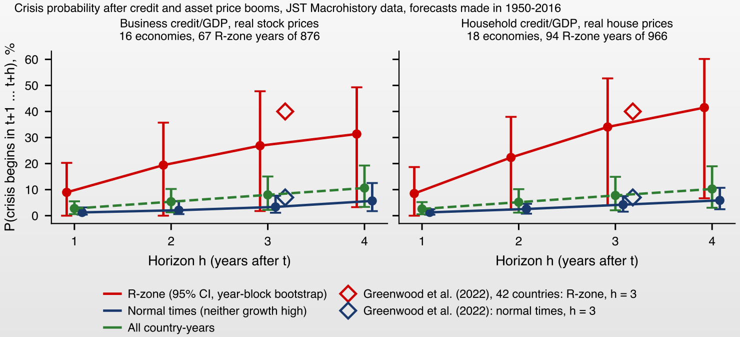
Case Study (5/5): Limits and Open Questions

- Data limit of our illustration
  - The public series are national: volume covers new homes, Case–Shiller the prices of existing homes (repeat sales)
  - The different universes limit the price–volume comparison
  - Short-term and non-occupant volume cannot be built
  - Figures 1–2 can be illustrated, Table 2 cannot
- Limits stated by the authors
  - Holding periods are realised, not intended (instrumental-variable check in the paper's Online Appendix)
  - The non-occupant proxy misclassifies owners who give the property address for tax mail
  - Selling horizons are exogenous in the model, so the counterfactuals may overstate short-term buyers
  - Credit and new construction are left out
- Open questions
  - Does trading volume also lead prices in crypto booms?
  - Is there a quiet (falling volume, rising prices) before the end of the Bitcoin episodes dated by BSADF?

Second Case Study (1/4): Are Financial Crises Predictable?

- ⊡ Paper: Greenwood, Hanson, Shleifer & Sørensen (2022), *Journal of Finance*, 77(2), 863–921
  - ▶ Setting: postwar systemic banking crises in 42 countries, 1950–2016
- ⊡ The question
  - ▶ Are financial crises unpredictable “bolts from the sky”?
  - ▶ Or do they predictably follow credit booms, as in the Kindleberger–Minsky sequence at the start of the chapter?
- ⊡ Why it matters for this chapter
  - ▶ The explosiveness tests use prices only; the paper adds the credit that finances the boom
  - ▶ A signal years before a crisis can inform macroprudential policy that leans against the wind
  - ▶ It does not justify it alone: the decision also needs the costs of intervening and of false alarms
- ⊡ Main claim (the paper’s abstract)
  - ▶ After 3 years of rapid credit and asset price growth, a crisis begins within the next 3 years with probability of about 40%
  - ▶ In normal times, with neither growing rapidly, the probability is about 7%
  - ▶ The authors call the combination of the two rapid rises the R-zone (red zone)

Second Case Study (2/4): Crisis Probability After the R-Zone



- ☐ Horizontal axis: horizon h in years; vertical axis: share of country-years t after which a crisis begins in $t+1, \ldots, t+h$
- ☐ Red: R-zone; blue: normal times; green: all country-years; diamonds: the numbers of the paper's abstract

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Second Case Study (3/4): Method and the Course Replication

- The paper's variables, for country i and year t (Eq. 2a–2c)
  - $\Delta_3(D/Y)_{it}$: 3-year change of credit to gross domestic product (GDP), in percentage points
  - $\Delta_3 \log P_{it}$: 3-year growth of the real price: stocks for firms, houses for households
  - R-zone: $\Delta_3(D/Y)$ above the 80th and $\Delta_3 \log P$ above the 66.7th percentile of the whole panel
  - $\mathrm{Crisis}_{i,t+1\to t+h} = 1$ if a crisis begins in country i in one of the years $t+1, \ldots, t+h$
- The paper: 42 countries, total credit, the crisis chronology of Baron, Verner & Xiong (2021)
  - Frequencies in 15 cells (price terciles × credit quintiles); in the R-zone cell, at 3 years: 45.3% for firms, 36.8% for households
  - Regressions with country fixed effects: Driscoll–Kraay errors, Kiefer–Vogelsang corrected p-values
  - Forecast origins stop in 2012, so every horizon has the same observations
- Course replication: the Jordà, Schularick & Taylor (2017) Macrohistory Database (JST), forecast origins 1950–2016, so up to 2016, not 2012
  - Firms: 16 economies; households: 18 economies; bank credit instead of total credit
  - JST chronology: 25 crises
  - R-zone thresholds, firms: credit +4.9 percentage points, real price +24.3 log points (about +27.5%)
  - Households: credit +6.3 percentage points, real price +12.7 log points (about +13.6%)
  - 95% confidence interval (CI): bootstrap over blocks of 5 calendar years, all countries of a year kept together
  - The thresholds are re-estimated in every replication; the CIs are pointwise, not simultaneous across sectors and horizons

Second Case Study (4/4): Result and Limits

- Households: after the R-zone, a crisis begins within 3 years in 34% of cases, CI [4%, 53%]
  - Normal times: 4%; the CI of the difference, [1.7, 46.2] percentage points, excludes zero
- Firms: 27%, CI [2%, 48%], against 3% in normal times
  - The CI of the difference, [0.1, 42.7] percentage points, barely excludes zero
- The significance depends on the block length
  - 3-year blocks: CIs of the difference [4.1, 46.0] (households) and [1.8, 40.8] (firms)
  - 8-year blocks: [−0.7, 44.3] and [−0.7, 41.8], both contain zero
- Crises are slow to arrive: after a household R-zone, 9% within one year and 41% within four
  - An R-zone occurs in the 3 years before 16 of the 25 crises (either sector)
- What the result supports: association
  - conditional frequencies, with no identified mechanism; the direction is the paper's, at a lower level
- Limits
  - Only 25 crises, many in 2007–2008: wide CIs, and a single global wave weighs heavily
  - The thresholds come from the whole sample, so they use future information; the paper also checks real-time thresholds

Can Bubble Alarms Warn of Crashes?

Real-time bubble alarms

- Questions
  - Can bubble alarms be raised in real time with a controlled false-alarm rate?
  - Do they carry crash information?
- BSADF date-stamping controls the size date by date only, not over the whole sample
  - Live research: monitoring rules for bubbles *and* crashes Whitehouse, Harvey & Leybourne (2023); "econometric thermometers" for housing Shi & Phillips (2023)
- Formal version: alarm $A_t \in \{0, 1\}$, crash $C_{t,h}$ = a fall of at least 20% within $h = 90$ days
  - $H_0 : P(C_{t,h} \mid A_t = 1) = P(C_{t,h} \mid A_t = 0)$; H_1: larger after an alarm
  - The criterion, fixed in advance: reject H_0 if the test robust to overlapping windows rejects it at 5%; otherwise, the data do not support the hypothesis
  - The absence of an effect is supported only by an equivalence test with a margin fixed in advance
- Identification Pástor & Veronesi (2009)
  - Explosive expected cash flows of a new technology leave the same signature as a bubble

The Discovery Loop and Its Checks

- **Literature**: ask for papers on the false-alarm rate of sequential bubble monitoring
  - Resolve every DOI (Digital Object Identifier) on Crossref; find the table the answer quotes
- **Hypotheses**: rational bubble, explosive expected cash flows, falling discount rate
  - Each with its own observable implication
  - E.g. an explosive price-to-earnings ratio vs explosive earnings
- **Code**: replicate a number from the chapter first, then extend
- **Robustness**: pre-register the grid of horizons and crash thresholds, with a Holm correction
  - Holm: the k-th smallest of m p-values is compared with $0.05/(m-k+1)$, stopping at the first non-rejection
- **Critique**: ask the assistant to attack the result as a referee
- **Checks** the researcher must run (▸ Appendix)
  - Full-sample information, the tail of the test
  - Changing volatility, multiplicity, overlapping outcomes

Mini-Case: Does the LPPLS Alarm Predict Crashes?

- A typical assistant proposal
  - "A positive LPPLS confidence indicator announces a crash; compare hit rates with Fisher's exact test"
  - Alarms come in runs and outcomes overlap, so we test by circular rotations of the alarm series
- One-sided p for $H_1 : \Delta > 0$ (higher hit rate after an alarm)
  - Bitcoin: $\Delta = -15.5$ percentage points, $p = 0.96$ (658 rotations); Nasdaq 100: $\Delta = -7.7$ points, $p = 0.98$
- Pre-registered grid of 4 horizons × 3 thresholds
  - Smallest raw p 0.76 (Bitcoin) and 0.69 (Nasdaq 100)
  - No cell rejects H_0 at 5%, before or after the Holm correction
- Interpretation (assumptions in the ▸ Appendix)
  - The indicator dates accelerating rises, but these tests do not support a higher probability of large falls after a signal
- What the result supports: association
  - tested between alarm and crash; no test of the grid finds it

Your Turn: a Project Seed

Questions

- Can a monitoring rule for Bitcoin, the Nasdaq 100 and NVIDIA keep the probability of any false alarm at 5% over the whole period?
- Can it still warn before large falls?

Replicate first

- GSADF = 3.85 for weekly Bitcoin
- The crash probability after run-ups above 100% in Greenwood, Shleifer & You (2019) (ours: 45%)

Extend

- Calibrate the alarm threshold by simulation over the whole monitoring period Whitehouse, Harvey & Leybourne (2023)
- Evaluate the alarms with rotations or a block bootstrap

Further Topics: deliverables, AI log

Key Takeaways

- A bubble is price minus fundamental value
    - Rational bubbles must be explosive; bursting bubbles hide from full-sample tests
- GSADF tests for explosive episodes
    - BSADF date-stamps them in real time
    - A late volatility rise distorts the size with Monte Carlo critical values; the wild bootstrap restores it
    - Date-stamping over many dates needs family-wise control
- Across our episodes the tests warned months before the peak, but did not time the crash
- LPPLS describes the acceleration
    - Its critical time moves with the window
    - The confidence indicator is judged by a dependence-robust test of hit rates
- Sharp run-ups raise crash risk without a statistically established average loss
    - A risk-management signal, not a trading rule

Which Tool When?

Question	**Tool**
Has there been any explosive episode?	GSADF with simulated critical values
When did it start and end?	BSADF date-stamping, minimum duration $\lceil \ln T \rceil$; family-wise critical values for long monitoring
Volatility changes a lot (crypto, crises)	Wild bootstrap critical values
Is the rise accelerating super-exponentially?	LPPLS confidence indicator over many windows
Has the risk state changed?	Markov-switching filtered probabilities
How likely is a crash after a run-up?	Run-up conditioning across many assets Greenwood, Shleifer & You (2019)

Check Yourself, and Next

Retrieval questions

- Why does a full-sample right-tailed ADF miss a bubble that collapses periodically?
- Why is the 95% critical value of GSADF far above that of a single ADF?
- Why is a single LPPLS critical time not a crash forecast?
- Why must hit rates after alarms be tested with a block bootstrap or rotations?

Chapter 18: systemic risk

- When a bubble bursts, losses spread through banks and markets: contagion and systemic risk
- How much one institution adds to the risk of the system
    - CoVaR: the value-at-risk (VaR) of the system when an institution is in distress
    - SRISK: the capital shortfall of an institution in a crisis
- Further reading: Franke, Härdle & Hafner (2019), Kindleberger & Aliber (2005), Shiller (2015), Sornette (2017)

The Practical Decision

- **What I choose**
  - for the existence of an episode: GSADF with wild-bootstrap critical values (Bitcoin: 3.85 against 2.82)
  - for dating: BSADF with the minimum duration $\lceil \ln T \rceil$ and a family-wise critical value for long monitoring
  - for crash risk: the crash frequency after sharp run-ups (45% after $+100\%$, CI $[27\%, 62\%]$), as a risk-management signal
  - the LPPLS alarm: the rule evaluated gives no robust evidence that it raises the probability of the crash defined here ($\Delta = -15.5$ percentage points for Bitcoin)
- **Under which conditions**
  - I read the test as dating explosive dynamics, not as proof of a bubble: explosive fundamentals leave the same footprint
  - an episode is confirmed only after L consecutive exceedances, so the real-time lead is shorter
- **What result would make me drop it**
  - an episode that fails the family-wise critical value, as Bitcoin 2020–2021 does
  - on a new sample, a CI of the crash frequency after run-ups above 100% that contains the 11% rate of a random industry-month

The Series Tested: Stocks, Industries, Crypto

Series	Frequency	Period	Use
S&P Composite, real price and dividend	monthly	1871–2026	price-dividend ratio
48 US industry portfolios	monthly	1926–2026	run-ups and crashes
Nasdaq 100, S&P 500	daily, monthly	1990–2026	dot-com, AI
BET, BET-FI	daily, monthly	1997–2026	Bucharest Stock Exchange indices
Bitcoin, Ethereum, Dogecoin	daily, weekly	2014–2026	crypto booms
NVIDIA, Cisco, Tesla, GameStop, Strategy (formerly MicroStrategy)	daily, monthly	1990–2026	single stocks

- Closing values up to 18 September 2026
  - Stocks adjusted for dividends and splits
  - Weekly and monthly series use the last price of the period
- Sources: shillerdata.com for the S&P Composite; Kenneth French Data Library for industries

The Dot-Com Bubble

pets.com
Sock Puppet

The Pets.com sock puppet, mascot of an online pet-supply retailer that listed and closed in 2000

Photo: Atomic Taco (2013); CC BY-SA 2.0; Wikimedia Commons

- Hedge funds rode the bubble and cut positions before the fall Brunnermeier & Nagel (2004)
- Yet the internet did change the economy: a technology revolution can create both value and a bubble Pástor & Veronesi (2009)

Levels Explode, Log Prices Accelerate

- Theory: the bubble is explosive in **levels**, $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$
  - The test regresses the **log** price
  - A pure bubble path $B_t = B_0(1+r)^t$ has a constant log growth $\ln(1+r)$: Δy_t does not depend on y_{t-1}, so $b = 0$
  - What $b > 0$ detects: a **rising** growth rate of $\ln P_t$, which occurs while the bubble share of $P_t = F_t + B_t$ grows

Appendix

- Toy example $(1 + r = 2)$: $F = 100$ constant, $B = 1, 2, 4, 8$, so $P = 101, 102, 104, 108$
  - Bubble share of the price: 1.0%, 2.0%, 3.8%, 7.4%
  - Log increments: $\ln(102/101) = 0.0099$, $\ln(104/102) = 0.0194$, $\ln(108/104) = 0.0377$
  - They rise with $y_{t-1} = \ln P_{t-1}$, so the fitted $b > 0$; the bubble alone, $B_t = 2^t$, has every increment equal to $\ln 2 = 0.693$, so $b = 0$
- Check: add $B = 16$, so $P = 116$
  - Is the next increment larger?
  - Yes, $\ln(116/108) = 0.0715$; once $B \gg F$ the increments approach $\ln 2$ and b falls back towards 0: windows that catch the **transition** have the power

Back

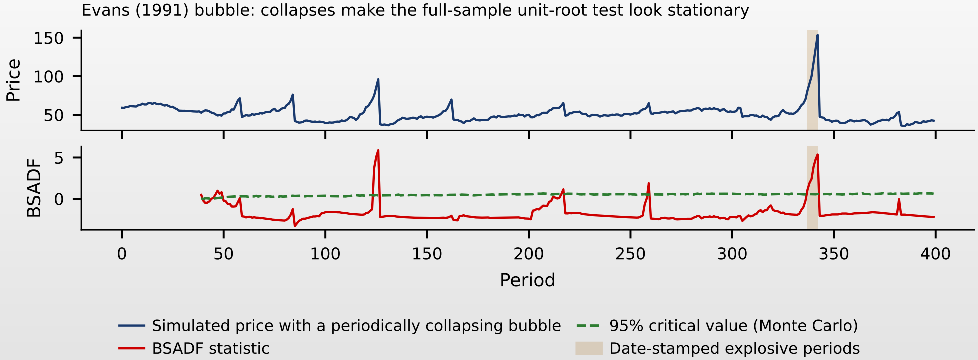
A Hand-Worked Date-Stamping Example

- Invented numbers; $L = 3$; critical value $cv_t = 1.0$ at every date
  - Flag $= 1$ if $\mathrm{BSADF}_t > cv_t$
  - s: start, c: confirmed, e: end, d: dropped

t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
BSADF_t	0.4	1.2	1.5	1.8	1.3	0.8	1.1	1.4	1.9	1.6	0.7	0.5	1.3	1.2	0.6
Flag	0	1	1	1	1	0	1	1	1	1	0	0	1	1	0
Event		s		c	e		s		c	e			d	d	

- Runs $t = 2$–5 and $t = 7$–10: four exceedances each, at least L, so both are kept
  - Confirmed at $t = 2 + L - 1 = 4$ and $t = 7 + L - 1 = 9$; each ends at its last exceedance, $t = 5$ and $t = 10$
  - Run $t = 13$–14: only two exceedances, fewer than L: dropped, never confirmed
- A toy chain links 2–5 and 7–10 (one quiet date, $t = 6$)
  - The chain allows at most one quiet date between runs
  - Episode containing $t = 10$: strict rule start $t = 7$; chain start $t = 2$, so the measured lead grows by five periods
- Check: flags 1 1 0 1 1 1 1 with $L = 3$
  - When is the first episode confirmed?
  - Answer: at $t = 6$; the run at $t = 1$–2 is too short

Evans Bubble: Full Sample vs Windows (1/2)



MFM_ch17_evans

Evans Bubble: Full Sample vs Windows (2/2)

- 400 simulated periods
  - BSADF against its 95% critical value from 2 000 random-walk paths
- Full-sample right-tailed ADF: -5.49 vs 95% critical value -0.09
  - No rejection, the bubble is invisible
- SADF 0.98 (critical value 1.46): no rejection
- GSADF 5.88 (critical value 2.21): the null is rejected
- BSADF crosses its critical value in several bursts
  - Only one lasts long enough to be date-stamped

MFM_ch17_evans

Size of GSADF When Volatility Changes: Design

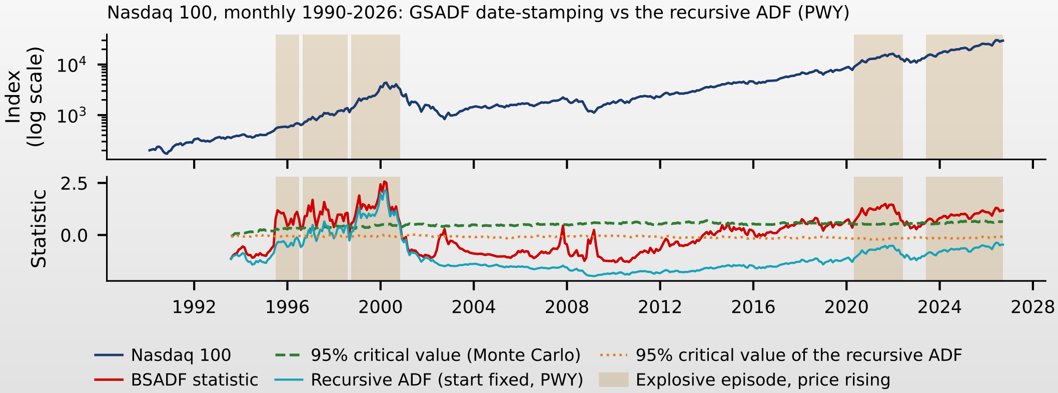
- Null model without a bubble: $y_t = T^{-1} + y_{t-1} + e_t$, a random walk with a small drift
- Four variance patterns of the mean-zero shocks e_t
  - Constant; standard deviation multiplied by 3 at $T/2$, so the variance by 9; standard deviation divided by 3 at $T/2$
  - GARCH (Generalised AutoRegressive Conditional Heteroskedasticity)(1,1): $\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2$, with $\alpha = 0.10$, $\beta = 0.88$
- Sample: $T = 400$, minimum window $w_0 = \lfloor r_0 T \rfloor = 40$ observations
  - 1 000 paths per scenario, each with its own wild bootstrap (199 replications)
  - Monte Carlo error: ±1.4 percentage points near 5%
- Two outcomes per path
  - Size: GSADF above its 95% critical value (Monte Carlo, MC in the table, or wild bootstrap)
  - False episode: BSADF above its pointwise 95% value for 6 consecutive periods

Size of GSADF When Volatility Changes: Results

Null model, no bubble	**GSADF, MC crit. value**	**GSADF, wild crit. value**	**False episode, MC**	**False episode, wild**
Constant volatility	4.1%	3.9%	49.1%	51.9%
Volatility $\times 3$ at $T/2$	32.0%	4.1%	63.8%	49.6%
Volatility $\times 1/3$ at $T/2$	4.3%	6.1%	48.0%	54.8%
GARCH(1,1), $\alpha = 0.10$, $\beta = 0.88$	16.6%	6.8%	58.1%	52.8%

- A late rise in volatility is mistaken for explosiveness Harvey et al. (2016)
  - Under GARCH the wild bootstrap removes most, not all, of the distortion; a fall in volatility is harmless
- Even with constant volatility, 49.1% of the paths contain a false date-stamped episode
  - The size is controlled date by date, not over the whole sample

Nasdaq 100: GSADF vs PWY Date-Stamping (1/2)



MFM_ch17_nasdaq

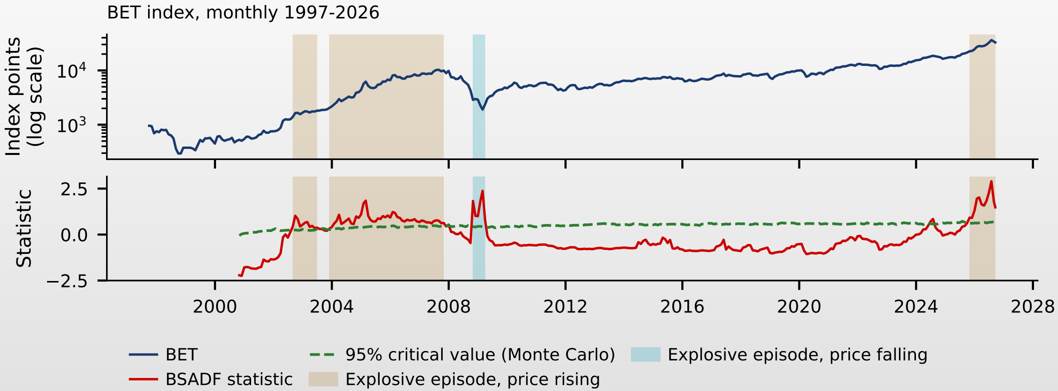
Back

Nasdaq 100: GSADF vs PWY Date-Stamping (2/2)

- Monthly log closes 1990–2026, $T = 440$, minimum window 42 months
- GSADF = 2.57 (95% critical value 2.19); SADF = 2.14 (1.41)
- BSADF episodes
  - Jun 1995–Jun 1996; Aug 1996–Jul 1998; Sep 1998–Oct 2000
  - Apr 2020–May 2022; May 2023–ongoing
  - PWY (fixed start) episodes: Apr 1997–Nov 1997; Jan 1998–Jul 1998; Sep 1998–Oct 2000
  - The recursive ADF misses the post-2020 episodes: it starts in 1990 and is dragged down by the long flat period after 2000
- Dot-com episode dated from Jun 1995: the start of a chain of 3 BSADF runs
  - 57 months before the peak of Mar 2000; confirmed in real time in Dec 1995

MFM_ch17_nasdaq

BET: The 2007 Boom and 2025 (1/2)



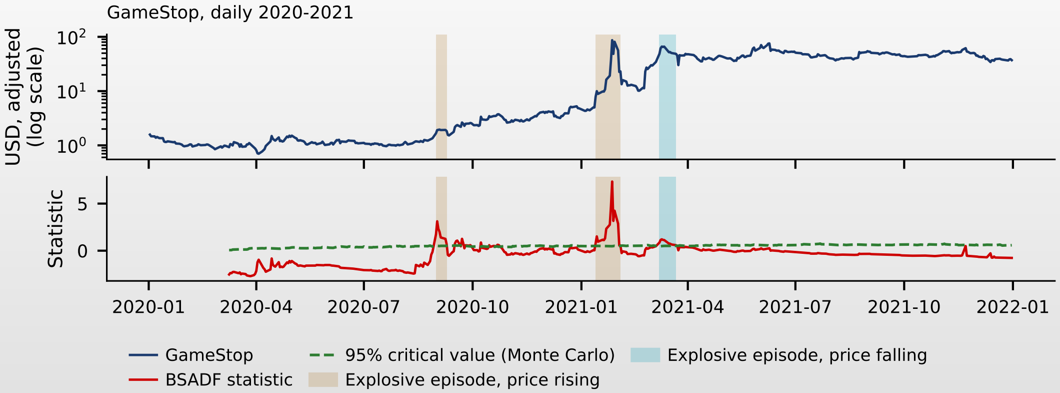
MFM_ch17_bet_gamestop

BET: The 2007 Boom and 2025 (2/2)

- Monthly log closes of the official BET, 1997–2026
  - $T = 348$, minimum window 37 months
- GSADF = 2.90 vs 95% critical value 2.21
  - Episodes: Aug 2002–Jun 2003; Nov 2003–Oct 2007; Oct 2008–Mar 2009; Oct 2025–ongoing
  - With one lag of the (autocorrelated) returns: GSADF = 2.29, below the critical value 2.39
- 2003–2007: the run-up around Romania's accession to the European Union (January 2007)
- Oct 2008–Mar 2009: an accelerating **fall**
- The current episode (from Oct 2025) is still open
  - Explosive does not mean it will crash: the rise is accelerating

MFM_ch17_bet_gamestop

GameStop, January 2021 (1/2)



MFM_ch17_bet_gamestop

GameStop, January 2021 (2/2)

A GameStop store, 3 February 2021, a week after the peak

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

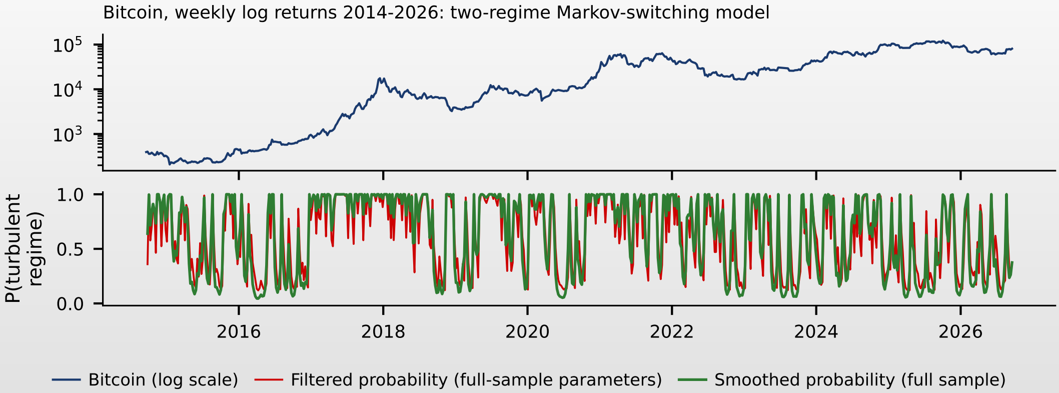
- Daily log closes 2020–2021, $T = 504$, window 45 days
- GSADF $= 7.34$, critical value 2.21
- January episode dated from 13 Jan 2021
  - 14 days before the closing high of 27 Jan 2021
- The episode is confirmed on 22 Jan 2021
- The chart dates two more short episodes: 31 Aug–9 Sep 2020 and 8–22 Mar 2021
- A surge of a few weeks: detecting it needs daily data
  - The test does not identify the mechanism (short covering: buying back shares sold short)

MFM_ch17_bet_gamestop

Calibration: Filimonov and Sornette

- Rewrite the cosine: $C\cos(\omega\ln(t_c - t) - \phi) = C_1\cos(\omega\ln(t_c - t)) + C_2\sin(\omega\ln(t_c - t))$ Filimonov & Sornette (2013)
  - $C_1 = C\cos\phi$, $C_2 = C\sin\phi$; A, B, C_1, C_2 are now **linear**: for given (t_c, m, ω) they come from OLS on the window $[t_1, t_2]$
- Search only over (t_c, m, ω)
  - A grid, then a Nelder–Mead refinement (a derivative-free simplex search) of the sum of squared residuals
- Search space Shu & Zhu (2020a, eq. 11)
  - $m \in [0, 1]$, $\omega \in [1, 50]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$
  - Damping $\frac{m|B|}{\omega|C|} \geq 1$: an approximate condition for a non-negative crash hazard rate
  - Exact condition: $m|B| \geq |C|\sqrt{m^2 + \omega^2}$
- Many local minima remain: different windows and starting points give different t_c Geraskin & Fantazzini (2013)

Bitcoin: Two Regimes (1/2)



MFM_ch17_markov_switching

Back

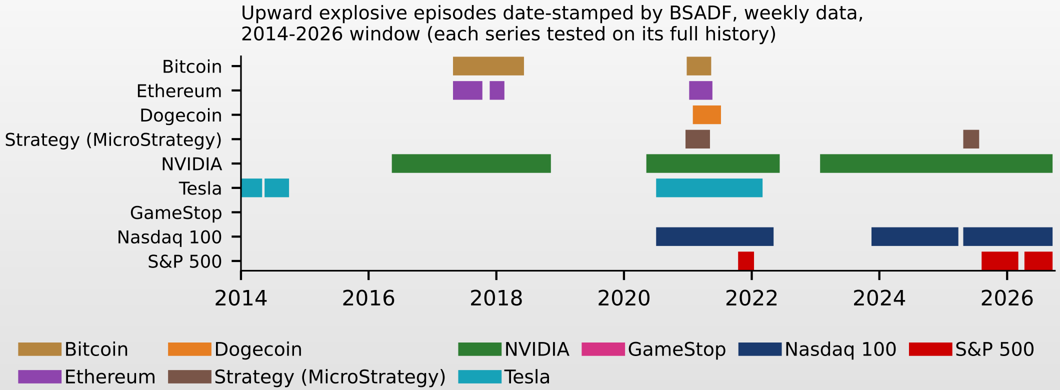
Bitcoin: Two Regimes (2/2)

Regime	$\hat{\mu}$ (SE)	$\hat{\sigma}$	$\hat{p}_{ii}$	Duration
Calm	0.26 (0.28)	2.93	0.736	3.8
Turbulent	1.20 (0.58)	11.22	0.840	6.3

- Weekly log returns in %, 2014–2026, 626 weeks
  - Duration in weeks; the turbulent regime covers 62% of the weeks
- Unlike equities, the turbulent regime has the **higher** mean
  - Regimes are short: volatility states, not bubble phases
- Regimes ignore the price level: a slow, calm bubble stays calm
  - BSADF says “the price is accelerating”; the regime says “risk has changed”

MFM_ch17_markov_switching

Explosive Episodes in Crypto and Tech Stocks (1/2)



MFM_ch17_crypto_ai

Explosive Episodes in Crypto and Tech Stocks (2/2)

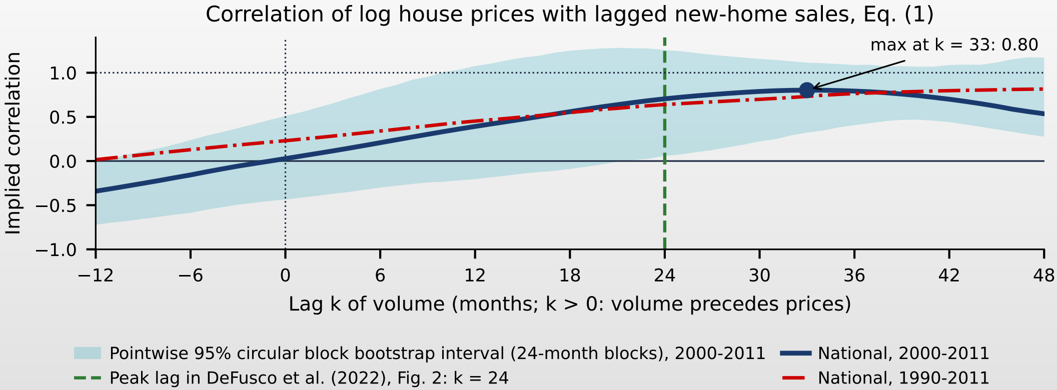
- Weekly log closes; rising episodes only
  - Each series tested on its own history, with its own Monte Carlo critical values
- Crypto booms cluster in 2017 and late 2020–2021
  - Bitcoin Apr 2017–Jun 2018; Dec 2020–May 2021; Ethereum Apr 2017–Oct 2017; Nov 2017–Feb 2018; Jan 2021–May 2021; Dogecoin Jan 2021–Jul 2021
- Stocks
  - Strategy (formerly MicroStrategy; holds Bitcoin): Dec 2020–May 2021; Apr 2025–Jul 2025; Tesla: Nov 2013–May 2014; May 2014–Oct 2014; Jul 2020–Mar 2022; NVIDIA: May 2016–Nov 2018; May 2020–Jun 2022; Jan 2023–ongoing
  - GameStop, weekly: the 3-week surge of January 2021 is shorter than the minimum duration
  - Indices: Nasdaq 100 Jul 2020–May 2022; Nov 2023–Mar 2025; Apr 2025–ongoing; S&P 500 Oct 2021–Jan 2022; Aug 2025–Mar 2026; Apr 2026–ongoing

MFM_ch17_crypto_ai

Case Study (1/3): Why It Matters, Data and Method

- The explosiveness tests of this chapter use prices only, yet bubbles come with high turnover Scheinkman & Xiong (2003)
  - Investors trade the resale option of Bubbles Beyond Rational Expectations
  - Quantities show who drives a boom, as blockchain flows did for the 2017 Bitcoin boom Griffin & Shams (2020)
- Data: 51 580 408 home sales from deeds records, 1995–2014, 115 metropolitan areas
  - **Speculators**: short-term buyers who resell within three years, and non-occupant buyers
- Lead–lag (Eq. 1): $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$, $k = -12, \ldots, 48$ months
  - i: metropolitan area; t: month; p: demeaned log price; v: demeaned sales rate; $\eta_{i,t}$: error
  - $k > 0$: volume k months earlier; implied correlation $\beta_k \, \mathrm{sd}(v_{i,t-k})/\mathrm{sd}(p_{i,t})$, sd: standard deviation

Case Study (2/3): Volume Leads Prices



MFM_ch17_case_study

Case Study (3/3): Volume Leads Prices

- The lead–lag regression (Eq. 1) on the national series, without a constant
  - Pointwise 95% band: circular block bootstrap, 24-month blocks, 2000 replications
- The implied correlation is a standardised slope, not a Pearson correlation: its bounds can exceed 1
- 2000–2011: maximum at $k = 33$ (0.80, CI $[0.32, 1.11]$)
  - At the paper's $k = 24$: 0.70, CI $[0.06, 1.26]$
- The band excludes zero for $k = 22, \dots, 48$
  - But one national cycle holds few independent blocks: wide intervals
- Direction as in the paper: volume first, by two to three years
  - On 1990–2011 the maximum falls at the edge of the grid ($k = 48$, 0.82): the lag is not identified

MFM_ch17_case_study

Your Turn: a Project Seed

- **Deliverables**
  - A pre-registered grid, fixed in a dated document before the evaluation
  - A notebook that regenerates every number; false-alarm and hit rates with valid *p*-values
- **AI log** (templates)
  - `AI_USE.md`: the prompts that shaped the design
  - `AI_ERRORS.md`: each wrong claim (left-tail critical values, a full-sample flag) and the test that caught it

Transversality: Where the Bubble Term Comes From

- Substitute the pricing equation into itself h times (law of iterated expectations)
  - $P_t = \sum_{i=1}^{h} \dfrac{\mathbb{E}_t[D_{t+i}]}{(1+r)^i} + \dfrac{\mathbb{E}_t[P_{t+h}]}{(1+r)^h}$
- Let $h \to \infty$: $P_t = F_t + B_t$ with $B_t = \lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h$
- **Transversality condition**: $\lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$
  - It holds $\Leftrightarrow B_t = 0 \Leftrightarrow P_t = F_t$
- Shift the limit one period: $\mathbb{E}_t[B_{t+1}] = \lim_h \mathbb{E}_t[P_{t+1+h}]/(1+r)^h = (1+r)\,B_t$
  - The condition that the discounted bubble is a martingale
- The condition $B_t \geq 0$ rules out a bubble that starts from zero Diba & Grossman (1988)
  - A bubble that is zero today stays zero

◀ Back

Bubbles for Fama: Effective Sample Size

- The run-ups fall in 32 calendar years and overlap
  - Consecutive 2-year windows of the same boom share months
- Year-block bootstrap SE: 8.9 percentage points
  - Binomial SE $\sqrt{\hat{p}(1-\hat{p})/n} = 6.8$, with $\hat{p} = 0.45$ and $n = 53$
- Design effect $\mathrm{deff} = (\mathrm{SE}_{\text{boot}}/\mathrm{SE}_{\text{bin}})^2 = 1.71$
  - Effective sample size $n_{\text{eff}} = n/\mathrm{deff} \approx 31$ independent observations
- Blocks of 2, 3 and 5 consecutive years: SE $= 9.0$, 8.6, 7.8
  - 2-year windows of adjacent years overlap
- Unconditional mean 2-year return over all industry-months: 28%
  - Above the CI of the mean return after a 100% run-up (-13% to 20%): returns after run-ups look lower than usual
  - A descriptive comparison, not a formal test of the difference of two dependent estimates

◂ Back

Why the Full-Sample ADF Fails: the Leverage of Collapses

- OLS slope: $\hat{b} = \sum_t \tilde{y}_{t-1}\,\Delta y_t \big/ \sum_t \tilde{y}_{t-1}^2$
  - $\tilde{y}_{t-1}$: the lagged log price minus its sample mean
  - Expansion: $\Delta y_t > 0$ while $\tilde{y}_{t-1}$ rises above its mean: positive terms
  - Collapse: the most negative Δy_t at the highest $\tilde{y}_{t-1}$: a few very large negative terms
- Evans form in logs, $y = \ln B$, collapse to $\delta = e^c$ with probability $1 - \pi$ Evans (1991)
  - For $B \gg \delta$ the survival log growth is $\approx \ln \frac{1+r}{\pi}$
  - $\mathbb{E}[\Delta y_t \mid y_{t-1}] \approx \pi \ln \frac{1+r}{\pi} + (1-\pi)(c - y_{t-1})$
  - Slope $-(1-\pi) < 0$: apparent mean reversion, although $\mathbb{E}_t B_{t+1} = (1+r)B_t$
- Windows with the expansion but not the collapse have $\hat{b} > 0$
  - When the log price accelerates (next slide)
  - A supremum over windows restores the power Phillips, Shi & Yu (2015b)

◂ Back

Explosive in Levels, Accelerating in Logs

- A rational bubble is explosive in the **level**: $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$
    - An autoregressive root $1 + r > 1$
- A deterministic example: F constant or growing more slowly than r, $B_t = (1+r)^t B_0$
    - The log price $\ln(F_t + B_t)$ **accelerates**: its growth rises from that of F towards $\ln(1+r)$, a convex path
    - A stochastic bubble does not guarantee this shape: for Blanchard–Watson, while it survives, the rate is $\ln((1+r)/\pi)$, and noise and collapses change the path
- So the log-price test targets a different alternative
    - **accelerating** log prices, not the level condition itself
    - The specification of the null and of the deterministic terms matters Phillips, Shi & Yu (2014)
- In this chapter we test log prices and log ratios (like PWY for the Nasdaq) Phillips, Wu & Yu (2011)
    - PSY test the price-dividend ratio Phillips, Shi & Yu (2015a)

◂ Back

Limit Theory (1): the Null Distribution

- Null model: $y_t = d\,T^{-\eta} + y_{t-1} + \varepsilon_t$, $\eta > 1/2$, ε_t iid $(0, \sigma^2)$ Phillips, Shi & Yu (2015a)
  - A random walk with an asymptotically negligible drift $d\,T^{-\eta}$ (d: a constant)
- Limit (PSY, Theorem 1, eq. 5), by the functional central limit theorem and continuous mapping
  - $$\mathrm{ADF}_{r_1}^{r_2} \Rightarrow \frac{\frac{1}{2} r_w \left[W(r_2)^2 - W(r_1)^2 - r_w\right] - \int_{r_1}^{r_2} W\,dr\,\left[W(r_2) - W(r_1)\right]}{r_w^{1/2}\left\{r_w \int_{r_1}^{r_2} W^2 dr - \left(\int_{r_1}^{r_2} W\,dr\right)^2\right\}^{1/2}}$$
  - $r_w = r_2 - r_1$; W: standard Brownian motion; $\Rightarrow$: convergence in distribution as $T \to \infty$
- σ and d disappear: the limit is pivotal
  - SADF and GSADF are suprema of the same functional: their critical values depend only on r_0 and the deterministic terms
- $r_1 = 0$, $r_2 = 1$: the Dickey–Fuller limit with a constant
  - Its 95% quantile is negative (simulated: -0.06), far from 1.645
  - The drift and intercept specification changes the limit and the size Phillips, Shi & Yu (2014)

◂ Back

Limit Theory (2): Mildly Explosive Alternatives

- ⊡ Alternative: $y_t = \rho_T y_{t-1} + \varepsilon_t$ with $\rho_T = 1 + c/k_T$, $c > 0$, $k_T \to \infty$, $k_T = o(T)$
  - ▶ PSY use $k_T = T^{\alpha}$, $\alpha \in (0, 1)$
  - ▶ c: how explosive; k_T: the speed at which the root approaches 1; $k_T = o(T)$: $k_T/T \to 0$
  - ▶ Closer to 1 than any fixed explosive root, farther than a local-to-unity root $1 + c/T$: the "mild" speed of real bubbles
- ⊡ Phillips and Magdalinos: $\dfrac{\rho_T^T}{\rho_T^2 - 1}(\hat{\rho}_T - \rho_T) \Rightarrow \mathcal{C}$ Phillips & Magdalinos (2007)
  - ▶ A standard Cauchy law, without assuming Gaussian errors
  - ▶ $\hat{\rho}_T$: the OLS estimate of ρ_T; $\mathcal{C}$: the standard Cauchy distribution
  - ▶ Rate $k_T \rho_T^T$: exponentially fast; the ADF statistic diverges to $+\infty$
- ⊡ Hence SADF and GSADF diverge too: the tests are consistent, power → 1 Phillips, Shi & Yu (2015b)
  - ▶ For GSADF any window of at least $r_0 T$ observations inside the bubble suffices
- ⊡ Finite samples: power depends on c, α and the duration
  - ▶ Short bursts may stay below the bar, as in the Evans simulation

◂ Back

Limit Theory (3): Consistent Date-Stamping

- Crossing times (PSY, eqs. 7–8)
  - $\hat{r}_e = \inf\{r_2 \geq r_0 : \mathrm{BSADF}_{r_2} > scv_{r_2}^{\beta_T}\}$, $\hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \mathrm{BSADF}_{r_2} < scv_{r_2}^{\beta_T}\}$
  - $\hat{r}_e$, $\hat{r}_f$: estimated start and end, as fractions of the sample; inf: the first r_2 that meets the condition
  - $scv_{r_2}^{\beta_T}$: the $100(1-\beta_T)\%$ quantile of BSADF on $\lfloor T r_2 \rfloor$ observations; β_T: the level; $\lfloor \cdot \rfloor$: rounded down
- Consistency (PSY, eq. 11): $(\hat{r}_e, \hat{r}_f) \to_p (r_e, r_f)$ if $\dfrac{1}{scv^{\beta_T}} + \dfrac{scv^{\beta_T}}{T^{1-\alpha/2}} \to 0$
  - $\to_p$: convergence in probability to the true dates r_e, r_f
  - The critical value must diverge, so that false alarms vanish ($\beta_T \to 0$)
  - But slower than $T^{1-\alpha/2}$, so that true bubbles are still found; e.g. like $\ln T$ Phillips, Shi & Yu (2015a)
- In practice β_T is fixed at 5% at every date
  - Over hundreds of dates a false alarm becomes likely (the Monte Carlo size study in the wild bootstrap section)

◂ Back

The Price-Dividend Ratio: Comparison with PSY

- Our critical values: 1 000 Monte Carlo paths of the sample length
- PSY also date 1879–1880, 1955–1956 and 1986–1987 Phillips, Shi & Yu (2015a)
  - On data to 2010 and without our filter of 8 consecutive months
- With a shorter minimum window 1879–1880, 1954–1956 and 2008–2009 appear in our data too

Back

Family-Wise Critical Values: Bootstrap Design

- Phillips and Shi: the 95% quantile of the maximum of BSADF over τ_b consecutive dates Phillips & Shi (2020)
    - Sets the FWER at 5% for the bootstrap design: τ_b monitoring dates after a start-up of w_0 observations
- Resampling
    - Residuals e_t of $\Delta y_t = a + e_t$, drawn with replacement and multiplied by $N(0,1)$ weights
    - Paths of $w_0 + \tau_b - 1$ changes; 999 replications
- Why the control is only approximate here
    - BSADF over the full history: later dates search more start points than the bootstrap design
    - Drawing with replacement loses the time order: unlike the pointwise wild bootstrap, which keeps each Δy_t at its date, these paths ignore the volatility clustering of Bitcoin
- The pointwise rule gives a crossing within the last 104 weeks in 42% of null paths

Back

Real Time vs Hindsight: Caveats

- Column definitions as on the main slide, with two caveats
  - Confirmed is only pseudo real time: w_0 and L come from the full-sample T (a truly real-time design fixes them ex ante)
  - GameStop: leads in calendar days converted to months

Back

Real-Time Monitoring: Newer Procedures

- Repeating a one-shot test every period inflates the false-positive rate Astill et al. (2018)
  - Astill et al. compare monitoring statistics with the maximum over a training period
  - The false-positive rate is then set for a chosen horizon, robust to time-varying volatility
- Bubbles and crashes Whitehouse, Harvey & Leybourne (2023)
  - After a bubble is detected, monitor for the crash with the sign of the mean of Δy_t
  - It never signals a crash before the crash starts
- Earlier detection Whitehouse, Harvey & Leybourne (2025)
  - Alternative variance standardisations keep the false-detection probability but find bubbles sooner (Bitcoin, housing)
- Collapse phase: PSY applied to the reversed sample dates the crisis and the recovery Phillips & Shi (2018)
- Alternatives to sup-ADF
  - A Chow-type break test, random walk to explosive, has the highest power in Homm & Breitung (2012)
  - Recursive tests with a start-point search Harvey, Leybourne & Sollis (2015)

Back

LPPL in Cryptocurrency and Emerging Markets

- Bitcoin 2012–2018: bubble episodes on several time scales, dated with LPPLS Gerlach, Demos & Sornette (2019)
- Bitcoin: real-time prediction of crashes with the LPPLS confidence indicator Shu & Zhu (2020b)
- Emerging markets in Latin America and Asia: log-periodic bubbles and anti-bubbles (accelerating declines) Johansen & Sornette (2001)
- Bucharest Stock Exchange
  - Herding and log-periodic patterns before the 2007–2008 crash Pele, Mazurencu & Nijkamp (2013)
  - Crash modelling for the same market Pele & Mazurencu-Marinescu (2012)
- Cryptocurrencies: a fundamental value from Metcalfe's law (network size)
  - Combined with LPPL bubbles Pele & Mazurencu-Marinescu-Pele (2019), Conda et al. (2026)
- Dogecoin 2021: LPPL together with sentiment from social media Pele (2025)
- Common lesson: LPPL describes the acceleration well
  - The crash date remains the weak point

◂ Back

Estimation: Hamilton Filter, Kim Smoother, EM

- Log-likelihood of θ (all μ_j, σ_j, p_{ij}), by prediction and update Hamilton (1989)
  - $\ell(\theta) = \sum_t \ln \sum_j P(s_t = j \mid \mathcal{F}_{t-1})\, \varphi(r_t; \mu_j, \sigma_j^2)$
  - $\mathcal{F}_t$: returns up to t; $\varphi(x; \mu, \sigma^2)$: the Normal density with mean μ and variance σ^2
- Kim smoother, backwards Kim (1994)
  - $P(s_t = j \mid \mathcal{F}_T) = P(s_t = j \mid \mathcal{F}_t) \sum_k p_{jk} \dfrac{P(s_{t+1} = k \mid \mathcal{F}_T)}{P(s_{t+1} = k \mid \mathcal{F}_t)}$
- EM algorithm: the M-step is a weighted average Hamilton (1990)
  - Smoothed $\xi_{tj} = P(s_t = j \mid \mathcal{F}_T)$, $\xi_{t,ij} = P(s_{t-1} = i, s_t = j \mid \mathcal{F}_T)$
  - $\hat{\mu}_j = \dfrac{\sum_t \xi_{tj} r_t}{\sum_t \xi_{tj}}, \quad \hat{\sigma}_j^2 = \dfrac{\sum_t \xi_{tj} (r_t - \hat{\mu}_j)^2}{\sum_t \xi_{tj}}, \quad \hat{p}_{ij} = \dfrac{\sum_{t \geq 2} \xi_{t,ij}}{\sum_{t \geq 2} \xi_{t-1,i}}$
- Unbounded likelihood ($\sigma_j \to 0$ on one return) and several local maxima
  - 20 random sets of starting values, then the full optimisation; if LR < 0, a refit with more starts
  - No penalty: a $\hat{p}_{ij} < 10^{-4}$ or a $\hat{\sigma}_j$ below 5% of the sample volatility flags a boundary maximum
  - Standard errors from the inverse numerical Hessian assume a known number of regimes

◂ Back

Why the LR Test for Regimes Is Non-Standard

- Under H_0 the regime means and variances coincide Hansen (1992)
  - p_{11}, p_{22} are not identified: nuisance parameters present only under H_1
  - The information matrix is singular and some scores are identically zero
- Hansen: bounds on the standardised LR via empirical-process theory Hansen (1992)
- Carrasco, Hu and Ploberger: an optimal test Carrasco, Hu & Ploberger (2014)
  - It needs only the estimate under H_0, with bootstrap critical values
- Two vs three regimes: the same logic
  - The null model is then the fitted two-regime chain

Back

Case Study, Data and Method (1/2)

- Data (Section 1.1): arm's-length sales only
  - 115 metropolitan areas (48% of the US housing stock); listings linked to sales in 57 of them
  - The cycle analysis covers 2000–2011: realised holding periods need consecutive sales
- Speculators (Section 1.2)
  - Short-term buyers: resell within three years (realised holding period)
  - Non-occupant buyers: a mailing address different from the property address (102 areas)
- Lead–lag (Eq. 1, Figure 2): construction of the variables
  - p: log price, demeaned by city; v: sales over the 2000 housing stock, demeaned by city and calendar month
  - Standard errors clustered by month

Back

Case Study, Data and Method (2/2)

- Cross-section (Table 2): OLS across cities on the speculative volume boom (2000–2005)
  - Outcomes: price boom (2000–2006), price bust (2006–2011), listings, volume, foreclosures
- Model (Sections 5–6): extrapolating buyers with different horizons
  - Not every listing sells
  - Counterfactuals: rational expectations, no speculators, transaction taxes

◀ Back

Case Study, Complete Results (1/2)

- Figure 4: sales rise from 2.73 to 3.82 million (2000 to 2005)
  - Short-term sales: from 510 to 940 thousand, 39% of the rise; non-occupant purchases: 53%
  - Interpretation: speculators account for much of the volume boom
- Table 2, Panel A: price-boom coefficient 1.930 (standard error 0.297, $R^2 = 0.272$)
  - One standard deviation more (12.9%, Table 1): price rise larger by 24.9 percentage points
  - Price bust -0.571 (0.083): 7.4 points deeper
  - Interpretation: more speculation, larger booms and larger busts

◂ Back

Case Study, Complete Results (2/2)

- Table 2, Panels B–C (Section 3.2)
  - Quantities in % of 2003 total volume
  - Unsold listings in the quiet: 5.961 (1.353), 76.9 points more per standard deviation
  - Foreclosures: 0.895 (0.398), 11.5 points, about 370 thousand more across the 115 areas
  - Interpretation: speculators who bought in the boom list in the quiet and fail to sell
  - That this turns the cycle is a model result (Sections 5–6), not identified by the regressions
- Figure 7, Panel C: a 10% price gain over the past year
  - Goes with an 8.2 points larger short-term buyer share
  - Interpretation: extrapolation draws short-horizon buyers in
- Table 6 (model): a tax of 5% of the purchase price, paid at sale
  - On all buyers: the simulated log-price bust falls from 8.2 to 1.1 log points (100× the change in log price)
  - On non-occupants only: to 5.8 points
  - Interpretation: short horizons, not non-occupancy, amplify the cycle

Back

Case Study, Lead–Lag Inference

- p and v demeaned over the full sample
- At $k = 0$: 0.03; at $k = -12$: -0.34
  - Level correlations, not a directional test of whether prices lead volume
- Eq. (1) has no constant and the lag-aligned series need not have zero mean
  - Hence bootstrap limits above 1 are possible
- The block bootstrap assumes stable, weakly dependent series
  - Doubtful for one boom–bust cycle in levels: the band is pointwise and exploratory

Back

The Discovery Loop, Reusable Prompts (1/2)

- **Literature**: map what is known, then verify every reference
  - ‘‘Which papers control the false-alarm rate of sequential bubble monitoring, and what size and power do they report? Give DOI and simulation design.’’
  - Resolve each DOI on Crossref and find the simulation table the answer quotes; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses**: competing explanations, each with its own observable
  - ‘‘For the current NVIDIA episode, list competing explanations (rational bubble, explosive expected cash flows, falling discount rate) and the observable each one predicts.’’
  - E.g. a bubble predicts an explosive price-to-earnings ratio; explosive cash flows predict explosive earnings with a stable ratio
- **Data and code**: on the course data, replicating a known number first
  - ‘‘Implement GSADF with r0 = 0.01 + 1.8/sqrt(T) on weekly Bitcoin closes from the course data and simulate its 95% critical value.’’
  - Replicate first: GSADF $= 3.85$ and critical value 2.27 on 626 weeks

Back

The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the alarms are evaluated
  - `‘‘Pre-register the evaluation of the alarm: horizons 30/60/90/180 days, crash thresholds 10/20/30%, inference robust to overlapping windows, Holm correction over the 12 cells.’’`
  - Twelve cells tried and one reported is data snooping; twelve cells pre-registered and corrected is a test
- **Critique**: ask the assistant to attack the result
  - `‘‘Act as a referee: why could an alarm-crash association be an artefact of overlapping windows, look-ahead or the choice of assets?’’`
  - Expect: dependent observations inflate significance; full-sample critical values; famous episodes only, the selection that Greenwood, Shleifer & You (2019) avoid
- **Report**: hit rates with valid p-values, all 12 cells, the alarms that failed
  - The prompts that shaped the design go into `AI_USE.md`

Back

What the Researcher Must Check

- **Full-sample information in date-stamping**
  - Critical values, lag order or detrending fitted on the whole sample
  - Test: cut the sample at t and recompute; the flag at t must not change
- **Wrong tail**
  - Left-tail ADF critical values (about -2.86) used against an explosive alternative
  - Test: on simulated random walks the right-tailed test must reject in about 5% of the paths
- **Changing volatility**
  - Test: wild bootstrap critical values; for Bitcoin 2.82 instead of 2.27
- **Multiplicity**: many dates, assets and windows
  - Test: run the rule on simulated random walks and count paths with at least one alarm
- **Overlapping outcomes**: Fisher's exact or χ^2 tests overstate significance
  - Test: block bootstrap or circular rotations of the alarm series; drop windows cut by the end of the data

◂ Back

Mini-Case: What the Result Does and Does Not Say

- A large one-sided p is not evidence of **no** information
  - It only fails to support a **higher** crash probability after alarms
  - Both point estimates are **negative**: one-sided rotation p for $\Delta < 0$, a lower crash rate after alarms, 0.07 (Bitcoin), 0.02 (Nasdaq 100)
  - Bitcoin: this 0.07 and the Newey–West one-sided 0.024 answer the same question under different null assumptions (stationary rotations vs a regression with HAC (Heteroskedasticity and Autocorrelation Consistent) errors)
  - Post-hoc, two-sided reading: alarms come in steady rises that did not end in a crash within 90 days
- Assumption behind the rotation p-values
  - Under H_0 every circular shift of the alarm series (at least 26 steps, ends joined) is as likely as the observed alignment
  - Stronger than equal hit rates: it needs stationary series and alarms independent of the crash process
  - Hence exploratory p-values; check their size by simulation under plausible null processes

Back

Further Study on Quantinar

Advanced courses on quantinar.com

- Cryptocurrency: Speculative Asset and Medium of Exchange
- Are Cryptos becoming alternative Assets?
- Statistics of Financial Markets

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

Back

Acronyms used in this material (1/2)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **BET**: Bucharest Exchange Trading (index)
- **BET-FI**: BET Financial Investment companies index
- **BSADF**: Backward Sup Augmented Dickey–Fuller (statistic)
- **CI**: Confidence Interval
- **CoVaR**: Conditional Value-at-Risk (of the system, given an institution)
- **DOI**: Digital Object Identifier
- **EM**: Expectation-Maximisation (algorithm)
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **GSADF**: Generalised Sup Augmented Dickey–Fuller (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- **JLS**: Johansen–Ledoit–Sornette (model)
- **JST**: Jordà–Schularick–Taylor (Macrohistory Database)
- **LPPL**: Log-Periodic Power Law
- **LPPLS**: Log-Periodic Power Law Singularity (model)
- **LR**: Likelihood Ratio (test)
- **MC**: Monte Carlo (simulation)
- **NVIDIA**: NVIDIA Corporation (company name, US chip designer)
- **OLS**: Ordinary Least Squares
- **PSY**: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- **PWY**: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- **SADF**: Sup Augmented Dickey–Fuller (test)
- **SE**: Standard Error
- **SRISK**: Systemic RISK (capital shortfall measure)
- **US**: United States
- **USD**: United States dollar
- **VaR**: Value-at-Risk

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